The Local-global Equivalence on General Networks: Applications*

Wonki Jo Cho† Changwoo Park‡

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Abstract

Strategy-proofness is an incentive property requiring immunity to preference misrepresentations. Local strategy-proofness, its weaker variant, requires immunity only to those manipulations that are local, or close, to the true preference type. Say that an environment satisfies the local-global equivalence (LGE) if each locally strategy-proof rule is indeed strategy-proof. Cho and Park (2025), our companion paper, consider environments that allow manipulation constraints to be asymmetric across types and characterize the environments satisfying LGE. Our objective is to provide applications that can be uniquely obtained by the approach in the latter work. The first application constructs an algorithm that enriches a given LGE-failing environment minimally so as to attain LGE. The second application shows how one can verify the relationship between strategy-proofness and some heuristic incentive properties proposed in the literature. Finally, the third application addresses the question of decomposing strategy-proofness into a combination of weaker properties.

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†Department of Economics, Korea University, Anam-ro 145, Seongbuk-gu, Seoul, 02841, South Korea; chowonki@korea.ac.kr

‡Department of Economics, University of Rochester, Rochester, NY, 14627, USA; changwoo.park@rochester.edu

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1 Introduction

In settings where agents report their preferences over alternatives, strategy-proofness says that a (preference) type should never benefit from reporting any other type admissible in the preference domain. In the presence of manipulation constraints that may not be uniform across types, one may weaken strategy-proofness to the requirement that no "local" (small) deviation is profitable. The resulting property is called local strategy-proofness. The "local-global equivalence" refers to the equivalence of strategy-proofness and its local variant. Starting with Carroll (2012) and Sato (2013), the LGE has been investigated in various contexts recently (Archer and Kleinberg, 2014; Cho, 2016; Mishra, 2016; Mishra et al., 2016; Kumar et al., 2021a, 2021b; Karmokar and Roy, 2022; Hong and Kim, 2023).

In our companion paper, Cho and Park (2025), we allow manipulation constraints to be asymmetric across types in that given any two types P and P', P may report P' but not necessarily the other way around. Such constraints can be captured by a directed network defined over types, which we call *environment*. In the network, an edge from a type to another indicates that the former can credibly misreport the latter. While the possibility of asymmetric misrepresentation has been extensively studied in the literature on mechanism design with partially verifiable information[1], it has not been addressed in the LGE context. Cho and Park (2025) characterize environments satisfying LGE; and they also identify conditions that are sufficient or necessary for environments satisfying a random version of LGE (RLGE). The objective of this paper is to put these results to work. We provide applications that can be obtained uniquely by the approach in Cho and Park (2025). Existing theorems on (R)LGE (Sato, 2013; Cho, 2016;

[1]See Green and Laffont (1986), Milgrom and Roberts (1986), Okuno-Fujiwara et al. (1990), Forges and Koessler (2005), and Bull and Watson (2007) among others.

Kumar et al., 2021a,b) cannot be applied because they only consider symmetry environments or their sufficient conditions for RLGE are too restrictive.

The first two applications of ours brings asymmetry of manipulation opportunities to the fore. In the first application, we begin with an arbitrary environment failing LGE and seek to add edges to it and restore LGE. Our algorithm called the *minimal bridge-building algorithm* produces an environment such that (i) it satisfies LGE; and (ii) removing any edge from it results in a violation of LGE. Construction of those "LGE-minimal" environments exploits the characterization of LGE in Cho and Park (2025): LGE is equivalent to strong connectedness, a property requiring that an environment be decomposed into various subenvironments, each of which is connected. The perspective underlying strong connectedness is distinct from Kumar et al. (2021a). Further, it can help identify, and fix, the part of a given environment attributing to the failure of LGE.

Our second application concerns verifying the relationship between strategy-proofness and some of its weakenings proposed in the literature. Altuntaş et al. (2023) assume that agents resort to a simple, heuristic operation when manipulating, which only drops a single alternative to the bottom of the preference list or pushes it up to the top. Their incentive properties, called *drop strategy-proofness* and *push-up strategy-proofness*, require immunity to such misrepresentations. Since dropping and pushing up induce asymmetric misrepresentation among types, their connection to strategy-proofness can easily be investigated with directed networks. Taking advantage of Cho and Park (2025), we reformulate drop strategy-proofness, push-up strategy-proofness, and their variants as different versions of local strategy-proofness and test the sufficiency of each of them for strategy-proofness. This significantly simplifies the traditional approach to verifying the sufficiency based on definitions.

Finally, in a rather unexpected application, we investigate the question of decomposing strategy-proofness into a combination of weaker properties. Roy and Sadhukhan (2022) identify domain conditions guaranteeing that strategy-proofness is equivalent to the combination of upper-contour strategy-proofness and some monotonicity properties. Applying our results on random LGE (RLGE)

in Cho and Park (2025), we show that RLGE underlies the latter equivalence, with an appropriately defined notion of local strategy-proofness; that is, under the relevant domain conditions, the combination of weaker properties implies strategy-proofness because it implies local strategy-proofness, which in turn implies strategy-proofness.

The rest of the paper is organized as follows. In Section 2, we set up the model and introduce conditions on environments related to LGE. Section 3 is our application to constructing LGE-minimal environments. Section 4 establishes the relationship between strategy-proofness and heuristic incentive properties. Section 5 shows how to apply the results on RLGE to decomposing strategy-proofness. All proofs are in Appendix A.

2 The Model

We consider (allocation) rules that take agents' report of their ordinal preferences over alternatives as input and return a lottery over those alternatives. Our focus is on the incentives of an individual agent to report truthfully. To ease exposition, we assume a single-agent setting. This assumption is without loss of generality because the incentive properties we study concern the relationship between an individual agent's report and his resulting welfare, taking all the other agent's reports as given.

Let $\boldsymbol{A}$ be a finite set of alternatives. The agent has a complete, transitive, and anti-symmetric preference relation $\boldsymbol{P}$ over A. Let $\boldsymbol{\mathcal{P}}$ be the set of all such preference relations. A **domain** is a non-empty subset $\boldsymbol{\mathcal{D}} \subseteq \mathcal{P}$. We call elements of $\mathcal{D}$ **types**. Assume that $|\mathcal{D}| \geq 3$. For all non-empty disjoint $B, C \subseteq A$, we write $\boldsymbol{B\ P\ C}$ if for each $b \in B$ and each $c \in C$, $b\ P\ c$. When B or C is a singleton, with $B = \{b\}$ or $C = \{c\}$, we simply write $b\ P\ C$ or $B\ P\ c$.

A (deterministic) **rule** is a single-valued mapping $\varphi : \mathcal{D} \to A$. It is **strategy-proof** if for all $P, P' \in \mathcal{D}$, $\varphi(P)\ P\ \varphi(P')$ or $\varphi(P) = \varphi(P')$. This definition relies on the idea that the agent can announce any type in the prespecified domain. In

other words, the agent has complete freedom to misrepresent his type. In practice, however, he may well face constraints when manipulating. We model such constraints by introducing a "network" defined over the domain. This approach was first proposed by Kumar et al. (2021a) using undirected networks, and later extended by Cho and Park (2025) to directed networks, which we adopt in this paper.

An **environment** is a pair $\langle \mathcal{D}, \mathcal{E} \rangle$, where $\mathcal{D}$ is a set of vertices (types) and $\mathcal{E} \subseteq \mathcal{D}^2$ is a set of *directed* edges. For all $P, P' \in \mathcal{D}$, the presence of an edge $(P, P') \in \mathcal{E}$ from P to P' means that type P can misrepresent himself as type P'. In figures, we represent $(P, P') \in \mathcal{E}$ by "$P \to P'$" (i.e., an arrow from P to P'); and if $(P, P') \in \mathcal{E}$ and $(P, P') \in \mathcal{E}$, we use "$P \longleftrightarrow P'$".

A rule φ is **locally strategy-proof** if for all $P, P' \in \mathcal{D}$ with $(P, P') \in \mathcal{E}$, $\varphi(P)\ P\ \varphi(P')$ or $\varphi(P) = \varphi(P')$. The following is our fundamental property of environments.

Definition 1. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies the **local-global equivalence (LGE)** if each locally strategy-proof rule on $\mathcal{D}$ is strategy-proof.

Next, we turn to "random" rules. A **lottery** $\lambda \equiv (\lambda_a)_{a \in A}$ is a probability distribution over A, where for each $a \in A$, λ_a is the probability of a being chosen and $\sum_{a \in A} \lambda_a = 1$. Let $\boldsymbol{\Delta(A)}$ be the set of lotteries over A. A **random rule** is a single-valued mapping $\varphi : \mathcal{D} \to \Delta(A)$. For each $P \in \mathcal{D}$ and each $B \subseteq A$, let $\boldsymbol{\varphi_B(P)} \equiv \sum_{b \in B} \varphi_b(P)$.

When a random rule is used, the issue arises of how the agent should evaluate lotteries over alternatives based solely on his ordinal preferences. We follow a prevalent approach in the literature, extending preferences over alternatives to preferences over lotteries by means of first-order stochastic dominance (Gibbard, 1977; Bogomolnaia and Moulin, 2001). Let $P \in \mathcal{D}$. For each $a \in A$, let $\boldsymbol{U(P, a)} \equiv \{b \in A : b\, P\, a\}$ be the (strict) **upper contour set of $\boldsymbol{P}$ at $\boldsymbol{a}$**. Also, for each $B \subseteq A$, B is an **upper contour set of $\boldsymbol{P}$** if for some $a \in A$, $U(P, a) = B$. Given $\lambda, \lambda' \in \Delta(A)$, we say that $\boldsymbol{\lambda}$ **(first-order) stochastically dominates $\boldsymbol{\lambda'}$ for $\boldsymbol{P}$**, denoted by $\boldsymbol{\lambda\ R^{sd}\ \lambda'}$, if for each upper contour set $U \subseteq A$ of P, $\sum_{a \in U} \lambda_a \geq \sum_{a \in U} \lambda'_a$. If, in addition,

the inequality is strict for some $a \in A$, $\boldsymbol{\lambda}$ **strictly stochastically dominates** $\boldsymbol{\lambda'}$ **for** $\boldsymbol{P}$ and we write $\boldsymbol{\lambda}\ \boldsymbol{P^{sd}}\ \boldsymbol{\lambda'}$.

A random rule φ is **strategy-proof** if for all $P, P' \in \mathcal{D}$, $\varphi(P)\ R^{sd}\ \varphi(P')$. It is **locally strategy-proof** if for all $P, P' \in \mathcal{D}$ with $(P, P') \in \mathcal{E}$, $\varphi(P)\ R^{sd}\ \varphi(P')$.

Definition 2. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies the **random local-global equivalence (RLGE)** if each locally strategy-proof random rule on $\mathcal{D}$ is strategy-proof.

2.1 Local-global Equivalence

Before presenting our results, we review existing findings on LGE and RLGE that our analysis exploits. Cho and Park (2025) identify two structural conditions on environments, each of which characterizes the environments satisfying LGE. We introduce these conditions in Section 2.1.1. Concerning RLGE, no complete characterization is currently available. However, Cho and Park (2025) discover a necessary condition and a sufficient condition. We discuss them in Section 2.1.2.

2.1.1 Local-global Equivalence for Deterministic Rules

A well-known necessary condition for LGE is that every type must be reachable from any other type via a sequence of directed edges in $\mathcal{E}$ (Sato, 2013; Kumar et al., 2021a). To introduce this condition, let $P, P' \in \mathcal{D}$. A **path from** $\boldsymbol{P}$ **to** $\boldsymbol{P'}$ is a sequence $\pi \equiv (P_1, \cdots, P_k)$ of distinct vertices in $\mathcal{D}$ such that $P_1 = P$, $P_k = P'$, and for each $\ell \in \{1, \cdots, k-1\}$, $(P_\ell, P_{\ell+1}) \in \mathcal{E}$. Let $\boldsymbol{\Pi(P, P')}$ be the set of all paths from P to P'. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ is **connected** if for all $P, P' \in \mathcal{D}$, $\Pi(P, P') \neq \emptyset$.

A **subdomain** is a non-empty subset of domain $\mathcal{D}$. For each subdomain $\mathcal{D}' \subseteq \mathcal{D}$, let $\boldsymbol{\mathcal{E}|_{\mathcal{D}'}} \equiv \{(P, P') \in \mathcal{E} : P, P' \in \mathcal{D}'\}$ be the restriction of $\mathcal{E}$ to the vertices in $\mathcal{D}'$. We refer to $\langle \mathcal{D}', \mathcal{E}|_{\mathcal{D}'} \rangle$ as a **subenvironment**. For all $P, P' \in \mathcal{D}$, a path $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P')$ is **in** $\boldsymbol{\langle \mathcal{D}', \mathcal{E}|_{\mathcal{D}'} \rangle}$ if $P_1, \cdots, P_k \in \mathcal{D}'$ and for each $\ell \in \{1, \cdots, k-1\}$, $(P_\ell, P_{\ell+1}) \in \mathcal{E}|_{\mathcal{D}'}$. Let $\boldsymbol{\Pi(P, P', \mathcal{E}|_{\mathcal{D}'})}$ be the set of all paths from P and P' in $\langle \mathcal{D}', \mathcal{E}|_{\mathcal{D}'} \rangle$. The subenvironment $\langle \mathcal{D}', \mathcal{E}|_{\mathcal{D}'} \rangle$ is **connected** if for all $P, P' \in \mathcal{D}'$, $\Pi(P, P', \mathcal{E}|_{\mathcal{D}'}) \neq \emptyset$. For each non-empty $B \subseteq A$ and each $a \in B$, let

$\boldsymbol{\mathcal{D}(a,B)} \equiv \{P \in \mathcal{D} : a\ P\ (B \setminus \{a\})\}$ be the set of all types in $\mathcal{D}$ that most prefers a in B. The following property, introduced by Cho and Park (2025), is necessary and sufficient for LGE.

Definition 3. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ is **strongly connected** if for each $B \subseteq A$ with $|B| \geq 2$ and each $a \in B$, the subenvironment $\langle \mathcal{D}(a,B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle$ is connected.

Next, we introduce another condition that characterizes the environments satisfying LGE. Let $P, P' \in \mathcal{D}$ and $a, b \in A$. We say that a **overtakes** b as we move from P to P' if $b\ P\ a$ and $a\ P'\ b$. Given a path $\pi = (P_1, \cdots, P_k) \in \Pi(P, P')$, the path is **with $\{a,b\}$-restoration** if for some $\ell_1, \ell_2, \ell_3 \in \{1, \cdots, k\}$ with $\ell_1 < \ell_2 < \ell_3$, $a\ P_{\ell_1}\ b$, $b\ P_{\ell_2}\ a$, and $a\ P_{\ell_3}\ b$. Otherwise, the path is **without $\{a,b\}$-restoration**. That is, overtaking between a and b occurs at most once along the path. For each $P \in \mathcal{D}$ and each $a \in A$, let $\boldsymbol{L(P,a)} \equiv \{b \in A : a\ P\ b\}$ be the (strict) **lower contour set of P at a**.

Definition 4. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies the **directed lower-contour-set no-restoration property**, or simply **Property DL**, if for all $P, P' \in \mathcal{D}$ and each $a \in A$, there is a path $\pi \in \Pi(P, P')$ such that for each $b \in L(P,a)$, π is without $\{a,b\}$-restoration.

The following result shows that LGE is characterized by each of strong connectedness and Property DL.

Theorem 1 (Cho and Park, 2025)**.** *For each environment $\langle \mathcal{D}, \mathcal{E} \rangle$, the following statements are equivalent:*

(1) *$\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies LGE.*

(2) *$\langle \mathcal{D}, \mathcal{E} \rangle$ is strongly connected.*

(3) *$\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies Property DL.*

2.1.2 Local-global Equivalence for Random Rules

We now turn to RLGE. Cho and Park (2025) refine the notion of strong connectedness and show that the resulting condition is necessary to ensure RLGE. To introduce this condition, for all non-empty $B, C \subseteq A$ with $|B| \geq 2$ and $C \subsetneq B$, let

$\boldsymbol{\mathcal{D}(C,B)} \equiv \{P \in \mathcal{D} : C\ P\ (B \setminus C)\}$ be the set of all types in $\mathcal{D}$, according to which each alternative in C is preferred to each alternative in $B \setminus C$.

Definition 5. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ is **hyper-connected** if for all non-empty $B, C \subseteq A$ with $|B| \geq 2$ and $C \subsetneq B$, $\langle \mathcal{D}(C,B), \mathcal{E}|_{\mathcal{D}(C,B)} \rangle$ is connected.

Theorem 2 (Cho and Park, 2025). *If an environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies RLGE, then it is hyper-connected.*

The converse of the theorem does not hold. To identify a sufficient condition for RLGE, Cho and Park (2025) refine Property DL. To introduce their refinement, let us first define the following notation and concepts. For $P \in \mathcal{D}$ satisfying, say, $C_1\ P\ C_2\ P\ x\ P\ \{y,z\}$, where $\{C_1, C_2, \{x\}, \{y,z\}\}$ is a partition of A, we simply write $\boldsymbol{P = [C_1, C_2, x, \{y,z\}]}$; similarly, we write the restriction $P|_{\{x,y,z\}}$ of P to a subset $\{x,y,z\}$ as $P|_{\{x,y,z\}} = [x, \{y,z\}]$. For all $P, P' \in \mathcal{D}$, there always exists a partition $\{C_1, C_2, \cdots, C_t\}$ of A such that $P = [C_1, C_2, \cdots, C_t]$ and $P' = [C_1, C_2, \cdots, C_t]$. The **associated partition for P and P'** (and for the edge (P, P') if $(P, P') \in \mathcal{E}$) is the finest of all partitions $\{C_1, C_2, \cdots, C_t\}$ with the above property.

For each $P \in \mathcal{D}$ and each $B \subseteq A$, let $\boldsymbol{L(P,B)} \equiv \{a \in A : B\ P\ a\}$ be the lower contour set of P at B. Clearly, if B is an upper contour set of P, then $L(P,B) = A \setminus B$.

Definition 6. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies the **upper-lower no-restoration property with limited overtaking**, or simply **Property ULLO**, if for all $P, P' \in \mathcal{D}$ and each upper contour set $B \subseteq A$ of P, there is a path $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P')$ such that (i) for each $b \in B$ and each $c \in L(P,B) = A \setminus B$, π is without $\{b,c\}$-restoration; and (ii) for each $\ell \in \{1, \cdots, k-1\}$, letting $\{C_1^\ell, \cdots, C_{t_\ell}^\ell\}$ be the associated partition for $(P_\ell, P_{\ell+1})$, for each $s \in \{1, \cdots, t_\ell\}$, $P_\ell|_{C_s^\ell} = [C_s^\ell \cap B, C_s^\ell \setminus B]$ or $P_{\ell+1}|_{C_s^\ell} = [C_s^\ell \setminus B, C_s^\ell \cap B]$.

Theorem 3 (Cho and Park, 2025). *If an environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies Property ULLO, then it satisfies RLGE.*

3 LGE-minimal Environments

In our first application, we start with an environment violating LGE and see how edges can be added to restore it. Taking advantage of the characterization of LGE in Theorem 1, we construct an algorithm that adds a minimal set of edges and achieves LGE. Due to its minimality nature, the question pursued here requires adopting directed networks as in our approach. Further, the workings of our algorithm hinge critically on the equivalence of LGE and strong connectedness and therefore emphasize tractability of strong connectedness.

3.1 The Simplest Case

Let us begin with the simplest setup where the domain is partitioned into two subdomains and types face different manipulation constraints depending on which subdomains they belong to. Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an environment. Suppose that there is a partition $\{\mathcal{D}_1, \mathcal{D}_2\}$ of $\mathcal{D}$ for which manipulation constraints are asymmetric for the types in $\mathcal{D}_1$ and $\mathcal{D}_2$ in the following sense:

Assumption 1. (1) *Each type in $\mathcal{D}_1$ can credibly report only the types in $\mathcal{D}_1$, so that for all $P, P' \in \mathcal{D}_1$ and each $P'' \in \mathcal{D}_2$, $(P, P') \in \mathcal{E}$ and $(P, P'') \notin \mathcal{E}$; and* (2) *each type in $\mathcal{D}_2$ faces no constraint, so that for each $P \in \mathcal{D}_2$ and each $P' \in \mathcal{D}$, $(P, P') \in \mathcal{E}$.*

It is not difficult to find contexts where manipulation is asymmetrically constrained in this manner. Assume that A is the set of public policies; and that $a, b \in A$ are respectively left-wing and right-wing policies. Let $\mathcal{D}_1 \equiv \mathcal{D}(a, \{a, b\})$ and $\mathcal{D}_2 \equiv \mathcal{D}(b, \{a, b\})$. A political candidate known to be from a leftist party belongs to $\mathcal{D}_1$ and can credibly report only the types in $\mathcal{D}_1$. On the other hand, an independent candidate that prefers b to a belongs to $\mathcal{D}_2$ and faces no constraint arising from party affiliation. Therefore, he can claim to be any type in $\mathcal{D}_1 \cup \mathcal{D}_2$. This gives rise to an environment satisfying Assumption 1.

Clearly, the environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfying Assumption 1 violates LGE. We ask how one can enrich $\mathcal{E}$, i.e., add edges to $\mathcal{E}$, minimally and restore LGE. This question, we find, is interesting in two respects. When a mechanism designer is

concerned with constructing a strategy-proof rule and $\langle \mathcal{D}, \mathcal{E} \rangle$ fails LGE, working with local manipulations allowed in $\mathcal{E}$ is insufficient. The designer may then take account of all manipulations directly to attain strategy-proofness but a more convenient avenue is to exploit the possibility that $\langle \mathcal{D}, \mathcal{E} \rangle$ can satisfy LGE if more edges are added to $\mathcal{E}$. Once LGE holds with an enriched set $\mathcal{E}^*$ of edges, it suffices for the designer to restrict attention to local manipulations in $\mathcal{E}^*$ for strategy-proofness. If, in addition, $\mathcal{E}^*$ is minimal among all sets of edges restoring LGE, the designer can efficiently identify exact edges required for LGE.

The question of constructing "LGE-minimal" environments from $\langle \mathcal{D}, \mathcal{E} \rangle$ also serves to illustrate strength of the approach and results by Cho and Park (2025). Unlike the existing approach, ours allow added edges to be directed. We can answer not only what types are involved in newly permitted manipulations but also what direction of manipulation (i.e., whether P misreports P' or the other way around) is truly required for the sake of LGE. Moreover, to provide a general algorithm constructing LGE-minimal environments, one needs a result that characterizes LGE by means of a computationally tractable property. The characterization of LGE (Theorem 1) involving strong connectedness can be used for the latter purpose. By contrast, it appears unclear from an algorithmic perspective how one can put the equivalence of LGE and Property DL to work.

Now we define LGE-minimality.

Definition 7. An environment $\langle \mathcal{D}, \mathcal{E}^* \rangle$ is **LGE-minimal relative to $\langle \mathcal{D}, \mathcal{E} \rangle$** if (i) $\mathcal{E} \subseteq \mathcal{E}^*$ and $\langle \mathcal{D}, \mathcal{E}^* \rangle$ satisfies LGE; and (ii) for each $\mathcal{E}'$ with $\mathcal{E} \subseteq \mathcal{E}' \subsetneq \mathcal{E}^*$, $\langle \mathcal{D}, \mathcal{E}' \rangle$ violates LGE.

Next, we introduce an algorithm that constructs LGE-minimal environments from $\langle \mathcal{D}, \mathcal{E} \rangle$. We call it the "minimal bridge-building algorithm" because in each of its steps, it adds an edge from $\mathcal{D}_1$ to $\mathcal{D}_2$ that serves as a "bridge" connecting $\mathcal{D}_1$ to $\mathcal{D}_2$.

The Minimal Bridge-building Algorithm. Let $\mathbf{\Phi}$ be the collection of (a, B)'s such that $B \subseteq A$ with $|B| \geq 2$, $a \in B$, $\mathcal{D}(a, B) \cap \mathcal{D}_1 \neq \emptyset$, and $\mathcal{D}(a, B) \cap$

$\mathcal{D}_2 \neq \emptyset$. By Assumption 1 on $(\mathcal{D}_1, \mathcal{D}_2)$, Φ is the collection of (a, B)'s for which $\mathcal{D}(a, B)$ is not connected.

Step 1: Construct a matrix $M \equiv \left(M_{(a,B),(P_1,P_2)}\right)_{(a,B)\in\Phi,(P_1,P_2)\in\mathcal{D}_1\times\mathcal{D}_2}$ of dimension $|\Phi| \times (|\mathcal{D}_1||\mathcal{D}_2|)$. Rows are indexed by $(a, B) \in\Phi$ and columns by $(P_1, P_2) \in \mathcal{D}_1 \times \mathcal{D}_2$. For each $(a, B) \in \Phi$ and each $(P_1, P_2) \in \mathcal{D}_1 \times \mathcal{D}_2$, $M_{(a,B),(P_1,P_2)} = 1$ if $P_1, P_2 \in \mathcal{D}(a, B)$ and 0 otherwise. For each column, calculate the column sum of its entries. Let $(P_{11}, P_{12}) \in \mathcal{D}_1 \times \mathcal{D}_2$ be a column with the largest column sum. Set $\mathcal{A}_1 \equiv \{(P_{11}, P_{12})\}$ ($\mathcal{A}_k$'s collect "added" edges) and $\Phi_1 \equiv \left\{(a, B) \in \Phi : M_{(a,B),(P_{11},P_{12})} = 1\right\}$.

Step k (where $k \geq 2$): Construct a submatrix M^k of M^{k-1} by removing column $(P_{k-1,1}, P_{k-1,2})$ and each row $(a, B) \in \Phi_{k-1}\backslash\Phi_{k-2}$ (take $M^{k-1} = M$ and $\Phi_{k-2} = \emptyset$ when $k = 2$). For each column of M^k, calculate the column sum of its entries. Let $(P_{k1}, P_{k2}) \in \mathcal{D}_1 \times \mathcal{D}_2$ be a column with the largest column sum. Let $\mathcal{A}_{k0} \equiv \mathcal{A}_{k-1} \cup \{(P_{k1}, P_{k2})\}$ and $\Phi_k \equiv \Phi_{k-1} \cup \left\{(a, B) \in \Phi\backslash\Phi_{k-1} : M^k_{(a,B),(P_{k1},P_{k2})} = 1\right\}$. Then proceed to the following Substeps $k.1$ through $k.(k-1)$.

Substep $k.\ell$ (where $\ell \in \{1, \cdots, k-1\}$): Using M, construct a vector $v \equiv \left(v_{(a,B)}\right)_{(a,B)\in\Phi_k}$ of dimension $|\Phi_k| \times 1$ such that for each $(a, B) \in \Phi_k$, $v_{(a,B)} = \sum_{(P_1,P_2)\in\mathcal{A}_{k,\ell-1}\backslash\{(P_{\ell1},P_{\ell2})\}} M_{(a,B),(P_1,P_2)}$. If $v \geqq \mathbf{1}$,[2] then set $\mathcal{A}_{k\ell} \equiv \mathcal{A}_{k,\ell-1}\backslash\{(P_{\ell1}, P_{\ell2})\}$; otherwise, set $\mathcal{A}_{k\ell} \equiv \mathcal{A}_{k,\ell-1}$.[3]

After completing Substep $k.(k-1)$, set $\mathcal{A}_k \equiv \mathcal{A}_{k,k-1}$.

The above process terminates after some Step T where $\Phi_T = \Phi$ (clearly, $T < \infty$ because $|\Phi| < \infty$ and for each $k \in \mathbb{N}$, $\Phi_k \subsetneq \Phi_{k+1}$). Let $\mathcal{A}_T$ be the set of edges obtained at the end of Step T. The final output of the algorithm is $\mathcal{E}^* \equiv \mathcal{E} \cup \mathcal{A}_T$. $\diamondsuit$

Example 1 below illustrates the minimal bridge-building algorithm. Our next result shows that $\langle\mathcal{D}, \mathcal{E}^*\rangle$, produced by the algorithm, is LGE-minimal relative to

[2]Here $\mathbf{1}$ denotes the vector of ones and the vector inequality "$v \geqq \mathbf{1}$" means that each entry of v is at least as large as 1.

[3]For $k = 2$, there is only one substep, Substep 2.1, and in that substep, the condition $v \geqq \mathbf{1}$ never holds. Thus, $\mathcal{A}_2 = \mathcal{A}_{21} = \mathcal{A}_{20}$.

$\langle \mathcal{D}, \mathcal{E} \rangle$.

Proposition 1. *Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an environment satisfying Assumption 1. Suppose that $\mathcal{E}^*$ is the set of edges obtained by applying the minimal bridge-building algorithm to $\langle \mathcal{D}, \mathcal{E} \rangle$. Then the environment $\langle \mathcal{D}, \mathcal{E}^* \rangle$ is LGE-minimal relative to $\langle \mathcal{D}, \mathcal{E} \rangle$.*

We omit the proof of this result as it is simpler than the proof of a more general result below, Proposition 2

3.2 The General Case

Assumption 1 on $\langle \mathcal{D}, \mathcal{E} \rangle$ can be relaxed in several directions. Before investigating the most general case, we first introduce some concepts that better explain our observations in Section 3.1, and then extend them to the general case.

Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an environment and $\{\mathcal{D}_1, \mathcal{D}_2\}$ a partition of $\mathcal{D}$. Consider an ordered pair $(j, k) \in \{1, 2\}^2$. For each subdomain $\mathcal{D}' \subseteq \mathcal{D}$, say that $\langle \mathcal{D}', \mathcal{E}|_{\mathcal{D}'} \rangle$ is **$(\mathcal{D}_j, \mathcal{D}_k)$-connected** if for each $P \in \mathcal{D}' \cap \mathcal{D}_j$ and each $P' \in \mathcal{D}' \cap \mathcal{D}_k$, $\Pi(P, P', \mathcal{E}|_{\mathcal{D}'}) \neq \emptyset$. Also, $\langle \mathcal{D}, \mathcal{E} \rangle$ is **$(\mathcal{D}_j, \mathcal{D}_k)$-strongly connected** if for each $B \subseteq A$ and each $a \in B$, $\langle \mathcal{D}(a, B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle$ is $(\mathcal{D}_j, \mathcal{D}_k)$-connected. It is simple to see how $(\mathcal{D}_j, \mathcal{D}_k)$-strong connectedness decomposes strong connectedness.

Lemma 1. *Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an environment and $\{\mathcal{D}_1, \mathcal{D}_2\}$ a partition of $\mathcal{D}$. Then $\langle \mathcal{D}, \mathcal{E} \rangle$ is strongly connected if and only if for each ordered pair $(j, k) \in \{1, 2\}^2$, $\langle \mathcal{D}, \mathcal{E} \rangle$ is $(\mathcal{D}_j, \mathcal{D}_k)$-strongly connected.*

With the new notion of connectedness above, we can state an assumption weaker than Assumption 1 in Section 3.1 as follows:

Assumption 2. *$\langle \mathcal{D}, \mathcal{E} \rangle$ is $(\mathcal{D}_1, \mathcal{D}_1)$-, $(\mathcal{D}_2, \mathcal{D}_2)$-, and $(\mathcal{D}_2, \mathcal{D}_1)$-strongly connected.*

Assumption 2 is weaker than Assumption 1 in several respects. An important difference is that Assumption 2 allows an arbitrary (possibly empty) set of edges from $\mathcal{D}_1$ to $\mathcal{D}_2$.[4] In view of Lemma 1, enriching $\mathcal{E}$ to obtain strong connectedness is equivalent to enriching $\mathcal{E}$ to obtain $(\mathcal{D}_1, \mathcal{D}_2)$-strong connectedness.

[4]Another difference is that for each $(j, k) \in \{(1, 1), (2, 1), (2, 2)\}$, $(\mathcal{D}_j, \mathcal{D}_k)$-strong connectedness is substantially weaker than the requirement that for each $P \in \mathcal{D}_j$ and each $P' \in \mathcal{D}_k$, $(P, P') \in \mathcal{E}$

Now we generalize the minimal bridge-building algorithm taking these issues into account.

The Minimal Bridge-building Algorithm (generalized). Let $\mathbf{\Phi}$ be the collection of (a, B)'s such that $B \subseteq A$ with $|B| \geq 2$, $a \in B$, $\mathcal{D}(a, B) \cap \mathcal{D}_1 \neq \emptyset$, and $\mathcal{D}(a, B) \cap \mathcal{D}_2 \neq \emptyset$. Construct a matrix $M \equiv \left(M_{(a,B),(P_1,P_2)}\right)_{(a,B)\in\Phi,(P_1,P_2)\in\mathcal{D}_1\times\mathcal{D}_2}$ of dimension $|\Phi| \times (|\mathcal{D}_1||\mathcal{D}_2|)$. Rows are indexed by $(a, B) \in \Phi$ and columns by $(P_1, P_2) \in \mathcal{D}_1 \times \mathcal{D}_2$. For each $(a, B) \in \Phi$ and each $(P_1, P_2) \in \mathcal{D}_1 \times \mathcal{D}_2$, $M_{(a,B),(P_1,P_2)} = 1$ if $P_1, P_2 \in \mathcal{D}(a, B)$ and 0 otherwise. Define

$$\mathcal{A}_0 \equiv \{(P_1, P_2) \in \mathcal{D}_1 \times \mathcal{D}_2 : (P_1, P_2) \in \mathcal{E}\}; \text{ and}$$
$$\Phi_0 \equiv \{(a, B) \in \Phi : \exists\, (P_1, P_2) \in \mathcal{A}_0 \text{ with } P_1, P_2 \in \mathcal{D}(a, B)\}.$$

Step 1: Construct a submatrix M^1 of M by removing each column $(P_1, P_2) \in \mathcal{A}_0$ and each row $(a, B) \in \Phi_0$. For each column of M^1, calculate the column sum of its entries. Let $(P_{11}, P_{12}) \in \mathcal{D}_1 \times \mathcal{D}_2$ be a column with the largest column sum. Set $\mathcal{A}_1 \equiv \mathcal{A}_0 \cup \{(P_{11}, P_{12})\}$ ($\mathcal{A}_k$'s collect "added" edges) and $\Phi_1 \equiv \Phi_0 \cup \left\{(a, B) \in \Phi\backslash\Phi_0 : M^1_{(a,B),(P_{11},P_{12})} = 1\right\}$.

Step k (where $k \geq 2$): Construct a submatrix M^k of M^{k-1} by removing column $(P_{k-1,1}, P_{k-1,2})$ and each row $(a, B) \in \Phi_{k-1}\backslash\Phi_{k-2}$. For each column of M^k, calculate the column sum of its entries. Let $(P_{k1}, P_{k2}) \in \mathcal{D}_1 \times \mathcal{D}_2$ be a column with the largest column sum. Let $\mathcal{A}_{k0} \equiv \mathcal{A}_{k-1} \cup \{(P_{k1}, P_{k2})\}$ and $\Phi_k \equiv \Phi_{k-1} \cup \left\{(a, B) \in \Phi\backslash\Phi_{k-1} : M^k_{(a,B),(P_{k1},P_{k2})} = 1\right\}$. Then proceed to the following Substeps $k.1$ through $k.(k-1)$.

Substep $k.\ell$ (where $\ell \in \{1, \cdots, k-1\}$): Using M, construct a vector $v \equiv \left(v_{(a,B)}\right)_{(a,B)\in\Phi_k}$ of dimension $|\Phi_k| \times 1$ such that for each $(a, B) \in \Phi_k$, $v_{(a,B)} = \sum_{(P_1,P_2)\in\mathcal{A}_{k,\ell-1}\backslash\{(P_{\ell1},P_{\ell2})\}} M_{(a,B),(P_1,P_2)}$. If $v \geqq \mathbf{1}$, then set $\mathcal{A}_{k\ell} \equiv \mathcal{A}_{k,\ell-1}\backslash\{(P_{\ell1}, P_{\ell2})\}$; otherwise, set $\mathcal{A}_{k\ell} \equiv \mathcal{A}_{k,\ell-1}$.[5]

After completing Substep $k.(k-1)$, set $\mathcal{A}_k \equiv \mathcal{A}_{k,k-1}$.

[5]For $k = 2$, there is only one substep, Substep 2.1, and in that substep, the condition $v \geqq \mathbf{1}$ never holds. Thus, $\mathcal{A}_2 = \mathcal{A}_{21} = \mathcal{A}_{20}$.

The above process terminates after some Step T where $\Phi_T = \Phi$ (clearly, $T < \infty$ because $|\Phi| < \infty$ and for each $k \in \mathbb{N}$, $\Phi_k \subsetneq \Phi_{k+1}$). Let $\mathcal{A}_T$ be the set of edges obtained at the end of Step T. The final output of the algorithm is $\mathcal{E}^* \equiv \mathcal{E} \cup \mathcal{A}_T$. $\diamondsuit$

The above minimal bridge-building algorithm adds only edges from $\mathcal{D}_1$ to $\mathcal{D}_2$, while keeping all the existing edges (such as those from $\mathcal{D}_2$ to $\mathcal{D}_1$). In fact, this results in a minimal enrichment of $\mathcal{E}$ satisfying $(\mathcal{D}_1, \mathcal{D}_2)$-strongly connectedness. Say that for each $(j,k) \in \{1,2\}^2$, $\langle \mathcal{D}, \hat{\mathcal{E}} \rangle$ is **minimal among all $(\mathcal{D}_j, \mathcal{D}_k)$-strongly connected environments containing $\mathcal{E}$** if (i) $\langle \mathcal{D}, \hat{\mathcal{E}} \rangle$ is $(\mathcal{D}_j, \mathcal{D}_k)$-strongly connected; and (ii) for each set $\mathcal{E}'$ of edges with $\mathcal{E} \subseteq \mathcal{E}' \subsetneq \mathcal{E}^*$, $\langle \mathcal{D}, \mathcal{E}' \rangle$ is not $(\mathcal{D}_j, \mathcal{D}_k)$-strongly connected.

Proposition 2. *Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an environment satisfying Assumption 2. Suppose that $\mathcal{E}^*$ is the set of edges obtained by applying the minimal bridge-building algorithm to $\langle \mathcal{D}, \mathcal{E} \rangle$. Then the environment $\langle \mathcal{D}, \mathcal{E}^* \rangle$ is minimal among all $(\mathcal{D}_1, \mathcal{D}_2)$-strongly connected environments containing $\mathcal{E}$. Thus, $\langle \mathcal{D}, \mathcal{E}^* \rangle$ is LGE-minimal relative to $\langle \mathcal{D}, \mathcal{E} \rangle$.*

If $(\mathcal{D}_2, \mathcal{D}_1)$-strong connectedness is dropped in Assumption 2, how can we construct an LGE-minimal environment relative to $\langle \mathcal{D}, \mathcal{E} \rangle$? By Lemma 1, we need to add edges to $\mathcal{E}$ appropriately to achieve $(\mathcal{D}_1, \mathcal{D}_2)$- and $(\mathcal{D}_2, \mathcal{D}_1)$-strong connectedness. By Proposition 2, the minimal bridge-building algorithm yields an environment $\langle \mathcal{D}, \mathcal{E}^* \rangle$ that is $(\mathcal{D}_1, \mathcal{D}_2)$-, $(\mathcal{D}_1, \mathcal{D}_1)$-, and $(\mathcal{D}_2, \mathcal{D}_2)$-strongly connected and is minimal among all such environments containing $\mathcal{E}$. Now we may switch the roles of $\mathcal{D}_1$ and $\mathcal{D}_2$ and apply the same algorithm to $\langle \mathcal{D}, \mathcal{E}^* \rangle$—which we henceforth refer to as **applying the algorithm to $(\mathcal{D}_2, \mathcal{D}_1)$**—to construct an environment $\langle \mathcal{D}, \mathcal{E}^{**} \rangle$ that achieves $(\mathcal{D}_2, \mathcal{D}_1)$-strong connectedness too. Then $\langle \mathcal{D}, \mathcal{E}^{**} \rangle$ is LGE-minimal relative to $\langle \mathcal{D}, \mathcal{E} \rangle$.

Now we are ready to drop Assumption 2 completely. Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an arbitrary environment violating LGE. Find a coarsest partition $\{\mathcal{D}_1, \cdots, \mathcal{D}_J\}$ of $\mathcal{D}$ such that for each $j \in \{1, \cdots, J\}$, $\langle \mathcal{D}, \mathcal{E} \rangle$ is $(\mathcal{D}_j, \mathcal{D}_j)$-strongly connected ($J \geq 2$ because LGE fails). Such coarsest partitions exist because we can always start with the partition whose cells are all singletons and make it coarser and coarser until

for some $j \in \{1, \cdots, J\}$, $(\mathcal{D}_j, \mathcal{D}_j)$-strong connectedness is violated. An argument similar to the proof of Lemma 1 shows that for strong connectedness, we should add edges to obtain $(\mathcal{D}_j, \mathcal{D}_k)$-strong connectedness for each $(j, k) \in \{1, \cdots, J\}^2$.

Once a coarsest partition $\{\mathcal{D}_1, \cdots, \mathcal{D}_J\}$ of $\mathcal{D}$ is identified, we can apply the minimal bridge-building algorithm to $(\mathcal{D}_1, \mathcal{D}_2)$ and then to $(\mathcal{D}_2, \mathcal{D}_1)$ (i.e., we add a minimal set of edges from $\mathcal{D}_1$ to $\mathcal{D}_2$ and then a minimal set of edges from $\mathcal{D}_2$ to $\mathcal{D}_1$). Denoting the resulting set of edges by $\mathcal{E}^1$, $\langle \mathcal{D}, \mathcal{E}^1 \rangle$ is minimal among all $(\mathcal{D}_1 \cup \mathcal{D}_2, \mathcal{D}_1 \cup \mathcal{D}_2)$-strongly connected[6] (and $(\mathcal{D}_j, \mathcal{D}_j)$-strongly connected for each $j \in \{3, \cdots, J\}$) environments containing $\mathcal{E}$. Now $\{\mathcal{D}_1 \cup \mathcal{D}_2, \mathcal{D}_3, \cdots, \mathcal{D}_J\}$ is a coarsest partition associated with $\langle \mathcal{D}, \mathcal{E}^1 \rangle$. We apply the minimal bridge-building algorithm to $(\mathcal{D}_1 \cup \mathcal{D}_2, \mathcal{D}_3)$ with the following modification. In each Step k, as an edge (P, P') from $\mathcal{D}_1 \cup \mathcal{D}_2$ to $\mathcal{D}_3$ is added to $\mathcal{E}^1$, we remove (i) each edge added in Steps 1 through $k-1$ that is made redundant by (P, P'); and (ii) each edge in $\mathcal{E}^1 \backslash \mathcal{E}$ that is made redundant by (P, P') in ensuring $(\mathcal{D}_1 \cup \mathcal{D}_2, \mathcal{D}_1 \cup \mathcal{D}_2)$-strong connectedness.[7] Next, we apply the latter algorithm to $(\mathcal{D}_3, \mathcal{D}_1 \cup \mathcal{D}_2)$. Letting $\mathcal{E}^2$ be the resulting set of edges, $\langle \mathcal{D}, \mathcal{E}^2 \rangle$ is minimal among all $(\mathcal{D}_1 \cup \mathcal{D}_2 \cup \mathcal{D}_3, \mathcal{D}_1 \cup \mathcal{D}_2 \cup \mathcal{D}_3)$-strongly connected environments containing $\mathcal{E}$. This process can be iterated until we get strong connectedness. Thus, we have established the following.

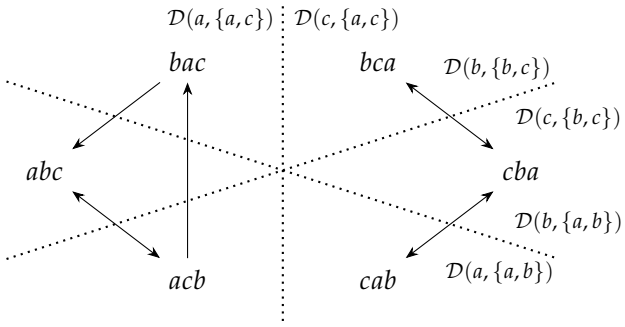
Theorem 4. *Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an environment failing LGE and $\{\mathcal{D}_1, \cdots, \mathcal{D}_J\}$ a coarsest partition of $\mathcal{D}$ associated with $\langle \mathcal{D}, \mathcal{E} \rangle$. Suppose that $\mathcal{E}^*$ is the set of edges obtained by applying the minimal bridge-building algorithm iteratively to $\{\mathcal{D}_1, \cdots, \mathcal{D}_J\}$. Then $\langle \mathcal{D}, \mathcal{E}^* \rangle$ is LGE-minimal relative to $\langle \mathcal{D}, \mathcal{E} \rangle$.*

We illustrate how the minimal bridge-building algorithm can be applied iteratively.

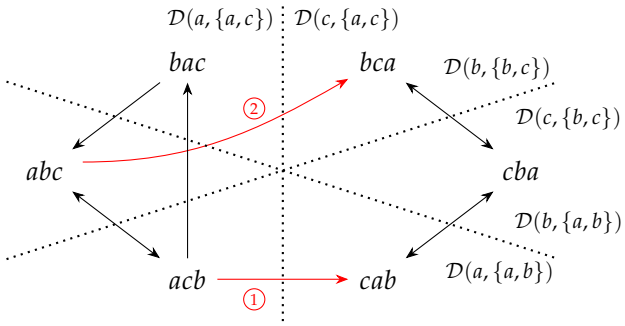
[6] In fact, the minimal bridge-building algorithm yields $(\mathcal{D}_1, \mathcal{D}_2)$- and $(\mathcal{D}_2, \mathcal{D}_1)$-strongly connectedness of $\langle \mathcal{D}, \mathcal{E}^1 \rangle$. Together with $(\mathcal{D}_1, \mathcal{D}_1)$- and $(\mathcal{D}_2, \mathcal{D}_2)$-strong connectedness, this implies $(\mathcal{D}_1 \cup \mathcal{D}_2, \mathcal{D}_1 \cup \mathcal{D}_2)$-strong connectedness.

[7] The substeps of Step k are designed to check whether adding an edge (P, P') makes any of the edges added in Steps 1 through $k-1$ redundant in ensuring $(\mathcal{D}_1 \cup \mathcal{D}_2, \mathcal{D}_3)$-connectedness of the subenvironments in $\left\{ \langle \mathcal{D}(a, B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle : (a, B) \in \Phi_{k-1} \right\}$. Similar to this, we should check whether adding (P, P') makes any edge in $\mathcal{E}^1 \backslash \mathcal{E}$ redundant in ensuring $(\mathcal{D}_1 \cup \mathcal{D}_2, \mathcal{D}_1 \cup \mathcal{D}_2)$-strong connectedness.

(a) Initial environment $\langle \mathcal{D}, \mathcal{E} \rangle$:



(b) Applying the MBB algorithm to $(\mathcal{D}_1, \mathcal{D}_2)$ yields $\langle \mathcal{D}, \mathcal{E}^1 \rangle$:

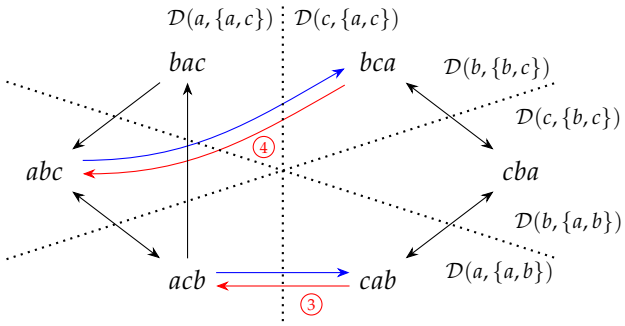


M^1	(acb, cab)	(abc, bca)	(abc, cab)	$\cdots$
$(a, \{a,b\})$	1	0	1	$\cdots$
$(b, \{b,c\})$	0	1	0	$\cdots$
$(c, \{b,c\})$	1	0	0	$\cdots$

$\downarrow$

M^2	(abc, bca)	(abc, cab)	$\cdots$
$(b, \{b,c\})$	1	0	$\cdots$

(c) Applying the MBB algorithm to $(\mathcal{D}_2, \mathcal{D}_1)$ yields $\langle \mathcal{D}, \mathcal{E}^2 \rangle$:

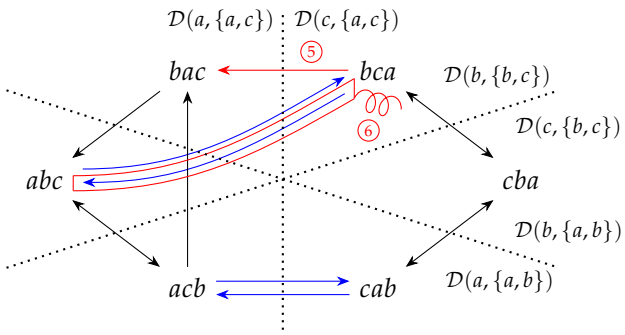


M^1	(cab, acb)	(bca, abc)	(cab, abc)	$\cdots$
$(a, \{a,b\})$	1	0	1	$\cdots$
$(b, \{b,c\})$	0	1	0	$\cdots$
$(c, \{b,c\})$	1	0	0	$\cdots$

$\downarrow$

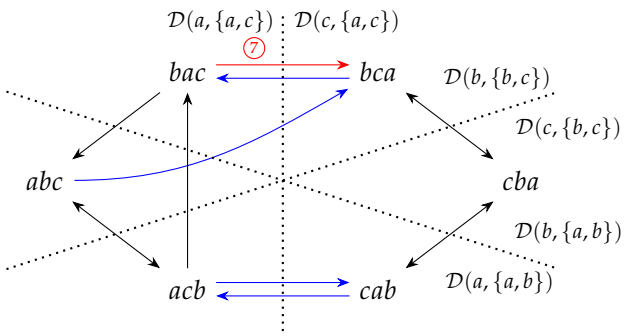
M^2	(bca, abc)	(cab, abc)	$\cdots$
$(b, \{b,c\})$	1	0	$\cdots$

(d) Applying the MBB algorithm to $(\mathcal{D}_1 \cup \mathcal{D}_2, \mathcal{D}_3)$ yields $\langle \mathcal{D}, \mathcal{E}^3 \rangle$:



M^1	(bca, bac)	(abc, bac)	(cba, bac)	$\cdots$
$(b, \{a,b\})$	1	0	1	$\cdots$
$(b, \{b,c\})$	1	1	0	$\cdots$
(b, A)	1	0	0	$\cdots$

(e) Applying the MBB algorithm to $(\mathcal{D}_3, \mathcal{D}_1 \cup \mathcal{D}_2)$ yields $\langle \mathcal{D}, \mathcal{E}^4 \rangle$:



M^1	(bac, bca)	(bac, cba)	(bac, acb)	$\cdots$
$(b, \{a,b\})$	1	1	0	$\cdots$
$(b, \{b,c\})$	1	0	0	$\cdots$
(b, A)	1	0	0	$\cdots$

(f) $\langle \mathcal{D}, \mathcal{E}^4 \rangle$ is LGE-minimal relative to $\langle \mathcal{D}, \mathcal{E} \rangle$:

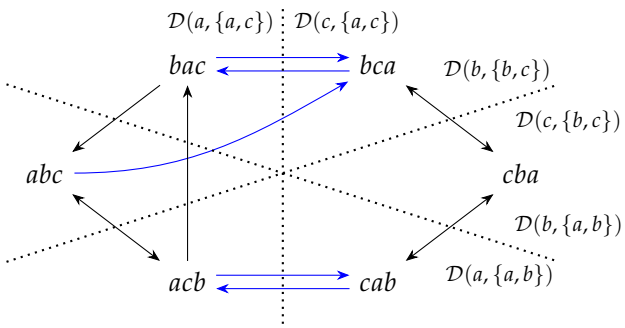


Figure 1: Illustration of the minimal bridge-building (MBB) algorithm (Example 1).

Example 1 (*Illustrating the minimal bridge-building algorithm*). Let $A \equiv \{a, b, c\}$ and consider an environment $\langle \mathcal{D}, \mathcal{E} \rangle$ in panel (a) of Figure 1, where $\mathcal{D}$ is the universal domain and $\mathcal{E}$ contains only the edges drawn in **black**. Clearly, $\langle \mathcal{D}, \mathcal{E} \rangle$ is not strongly connected. With $\mathcal{D}_1 \equiv \{abc, acb\}$, $\mathcal{D}_2 \equiv \{cab, cba, bca\}$, and $\mathcal{D}_3 \equiv \{bac\}$, $\{\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3\}$ is a coarsest partition associated with $\langle \mathcal{D}, \mathcal{E} \rangle$ and for each $j \in \{1, 2, 3\}$, $\langle \mathcal{D}, \mathcal{E} \rangle$ is $(\mathcal{D}_j, \mathcal{D}_j)$-strong connected.

Let us first apply the minimal bridge-building algorithm to $(\mathcal{D}_1, \mathcal{D}_2)$. One can check that $\Phi = \{(a, \{a, b\}), (b, \{b, c\}), (c, \{b, c\})\}$, $\mathcal{A}_0 = \emptyset$, and $\Phi_0 = \emptyset$. The algorithm adds edges to make the subenvironments associated with the pairs in $\Phi \backslash \Phi_0$ $(\mathcal{D}_1, \mathcal{D}_2)$-connected. The matrix M^1 obtained at the beginning of Step 1 is given in Figure 1.(b) (M^1 has six columns but we indicate several key ones only). Among all columns of M^1, the column (acb, cab) has the largest column sum of 2. Thus, we set $\mathcal{A}_1 \equiv \mathcal{A}_0 \cup \{(acb, cab)\}$ (i.e., add edge ① in Figure 1.(b) to $\mathcal{E}$) and $\Phi_1 \equiv \Phi_0 \cup \{(a, \{a, b\}), (c, \{b, c\})\}$. Next, in line with Step 2, we remove rows $(a, \{a, b\})$ and $(c, \{b, c\})$ and column (acb, cab) from M^1, to get M^2. Among all columns of M^2, the column (abc, bca) has the largest column sum of 1. Thus, we set $\mathcal{A}_2 \equiv \mathcal{A}_1 \cup \{(abc, bca)\}$ (i.e., add edge ② to $\mathcal{E} \cup \{(acb, cab)\}$) and $\Phi_2 \equiv \Phi_1 \cup \{(b, \{b, c\})\}$.[8] Since $\Phi_2 = \Phi$, the algorithm stops after Step 2 and its output is $\mathcal{E}^1 \equiv \mathcal{E} \cup \mathcal{A}_2 = \mathcal{E} \cup \{(acb, cab), (abc, bca)\}$ (in Figure 1.(b) and subsequent figures, **red** arrows indicate edges added in the present application of the minimal bridge-building algorithm, and **blue** arrows indicate edges added in previous applications). The environment $\langle \mathcal{D}, \mathcal{E}^1 \rangle$ is minimal among all $(\mathcal{D}_1, \mathcal{D}_2)$-strongly connected environments containing $\mathcal{E}$.

Next, we apply the minimal bridge-building algorithm to $(\mathcal{D}_2, \mathcal{D}_1)$ and achieve $(\mathcal{D}_2, \mathcal{D}_1)$-strong connectedness. Refer to Figure 1.(c). An argument symmetric to the one in the previous paragraph shows that the algorithm adds edges ③ and ④ to $\mathcal{E}^1$ and its output is $\mathcal{E}^2 \equiv \mathcal{E}^1 \cup \{(cab, acb), (bca, abc)\}$. The environment $\langle \mathcal{D}, \mathcal{E}^2 \rangle$ is minimal among all $(\mathcal{D}_1 \cup \mathcal{D}_2, \mathcal{D}_1 \cup \mathcal{D}_2)$-strongly connected environments containing $\mathcal{E}$.

With $\langle \mathcal{D}, \mathcal{E}^2 \rangle$, it remains to connect $\mathcal{D}_1 \cup \mathcal{D}_2$ and $\mathcal{D}_3$. Let us apply the min-

[8] In Step 2, substeps has no practical influence on the set of added edges; see Footnote 5.

imal bridge-building algorithm to $(\mathcal{D}_1 \cup \mathcal{D}_2, \mathcal{D}_3)$. As shown in Figure 1.(d), it follows that $\Phi = \{(b, \{a, b\}), (b, \{b, c\}), (a, \{a, c\}), (b, A)\}$, $\mathcal{A}_0 = \{(acb, bac)\}$, and $\Phi_0 = \{(a, \{a, c\})\}$, and the algorithm adds edges to make the subenvironments associated with the pairs in $\Phi \backslash \Phi_0$ $(\mathcal{D}_1 \cup \mathcal{D}_2, \mathcal{D}_3)$-connected. The column (bca, bac) in M^1 attains the largest column sum of 3. Thus, the edge (bca, bac), i.e., ⑤ in the figure, is added to $\mathcal{E}^2$. This, however, makes edge ④ from Figure 1.(c) redundant in keeping $(\mathcal{D}_1 \cup \mathcal{D}_2, \mathcal{D}_1 \cup \mathcal{D}_2)$-strongly connected; letting $\mathcal{E}^3 \equiv \left(\mathcal{E}^2 \cup \{(bca, bac)\}\right) \backslash \{(bca, abc)\}$, $\langle \mathcal{D}(b, \{b, c\}), \mathcal{E}^3|_{\mathcal{D}(b,\{b,c\})} \rangle$ is $(\mathcal{D}_1 \cup \mathcal{D}_2, \mathcal{D}_1 \cup \mathcal{D}_2)$-connected. Therefore, the last step involves removing edge ④, indicated as operation ⑥ in the figure. The algorithm stops at this point, yielding $\mathcal{E}^3$ as the output. The environment $\langle \mathcal{D}, \mathcal{E}^3 \rangle$ is minimal among all $(\mathcal{D}_1 \cup \mathcal{D}_2, \mathcal{D}_3)$- and $(\mathcal{D}_1 \cup \mathcal{D}_2, \mathcal{D}_1 \cup \mathcal{D}_2)$-strongly connected environments containing $\mathcal{E}$.

Finally, we apply the minimal bridge-building algorithm to $(\mathcal{D}_3, \mathcal{D}_1 \cup \mathcal{D}_2)$. See Figure 1.(e). With the same Φ as in the previous paragraph, the edge (bac, bca), i.e., ⑦ in the figure, is added to $\mathcal{E}^3$. The algorithm terminates here. Its output is $\mathcal{E}^4 \equiv \mathcal{E}^3 \cup \{(bac, bca)\}$ and $\langle \mathcal{D}, \mathcal{E}^4 \rangle$ is minimal among all $(\mathcal{D}_1 \cup \mathcal{D}_2 \cup \mathcal{D}_3, \mathcal{D}_1 \cup \mathcal{D}_2 \cup \mathcal{D}_3)$-strongly connected environments containing $\mathcal{E}$. Since $\mathcal{D}_1 \cup \mathcal{D}_2 \cup \mathcal{D}_3 = \mathcal{D}$, there is no further subdomains to connect by the algorithm. We conclude that $\langle \mathcal{D}, \mathcal{E}^4 \rangle$, given in Figure 1.(f), is LGE-minimal relative to $\langle \mathcal{D}, \mathcal{E} \rangle$. △

4 Incentive Properties Based on Heuristic Manipulations

There are some incentive properties that involve simple heuristic manipulations. In this section, we apply the results on (R)LGE, discussed in Section 2.1, to them and see whether they are equivalent to the stronger requirement of strategy-proofness.

When agents make decisions, they often turn to heuristic principles and simplify the otherwise complex task of comprehending consequences of their choices (Tversky and Kahneman, 1974; Kahneman et al., 1982; Gilovich et al., 2002). In

the manipulation context, those heuristic principles lead them to deviate from their true preference types through simple operations. Altuntaş et al. (2023) consider two heuristic manipulations of this nature: an agent may choose an alternative and drop it to the bottom of his preference rankings or push it up to the top. *Throughout this section, assume that $\mathcal{D}$ is the universal domain.* For all $P, P' \in \mathcal{D}$, P' is a **drop strategy** for P if there is $a \in A$ such that $A \backslash \{a\}\, P'\, a$ and $P'|_{A\backslash\{a\}} = P|_{A\backslash\{a\}}$; also, P' is a **push-up strategy** for P if there is $a \in A$ such that $a\, P'\, A \backslash \{a\}$ and $P'|_{A\backslash\{a\}} = P|_{A\backslash\{a\}}$. Incentive properties that restrict attention to these particularly simple manipulations are defined below.

Drop Strategy-proofness: For all $P, P' \in \mathcal{D}$ such that P' is a drop strategy for P, $\varphi(P)\, P\, \varphi(P')$ or $\varphi(P) = \varphi(P')$.

Push-up Strategy-proofness: For all $P, P' \in \mathcal{D}$ such that P' is a push-up strategy for P, $\varphi(P)\, P\, \varphi(P')$ or $\varphi(P) = \varphi(P')$.

If the agent drops or pushes up an alternative to misrepresent his preferences, it is also conceivable that he may engage in such operations multiple times consecutively, to form a report that further deviates from the truth. Let $k \in \mathbb{N}$. For all $P, P' \in \mathcal{D}$, P' is a ***k*-drop strategy** for P if P' is obtained from P by dropping k alternatives in a row, i.e., there is a sequence $(P_1, \cdots, P_{k+1})$ in $\mathcal{D}$ such that $P_1 = P$, $P_{k+1} = P'$, and for each $\ell \in \{1, \cdots, k\}$, $P_{\ell+1}$ is a drop strategy for P_ℓ; similarly, P' is a ***k*-push-up strategy** for P if P' is obtained from P by pushing up k alternatives in a row, i.e., there is a sequence $(P_1, \cdots, P_{k+1})$ in $\mathcal{D}$ such that $P_1 = P$, $P_{k+1} = P'$, and for each $\ell \in \{1, \cdots, k\}$, $P_{\ell+1}$ is a push-up strategy for P_ℓ. One way of strengthening drop and push-up strategy-proofness is to require that for any $\ell \in \{1, \cdots, k\}$, the agent should not gain from ℓ-drop and ℓ-push-up strategies.

***k*-Drop Strategy-proofness:** For all $P, P' \in \mathcal{D}$ such that for some $\ell \in \{1, \cdots, k\}$, P' is an ℓ-drop strategy for P, $\varphi(P)\, P\, \varphi(P')$ or $\varphi(P) = \varphi(P')$.

***k*-Push-up Strategy-proofness:** For all $P, P' \in \mathcal{D}$ such that for some $\ell \in \{1, \cdots, k\}$, P' is an ℓ-push-up strategy for P, $\varphi(P)\, P\, \varphi(P')$ or $\varphi(P) = \varphi(P')$.

If the rule under consideration is random, then k-drop and k-push-up strategy-proofness would require "$\varphi(P)\, R^{sd}\, \varphi(P')$" instead of "$\varphi(P)\, P\, \varphi(P')$ or $\varphi(P) =$

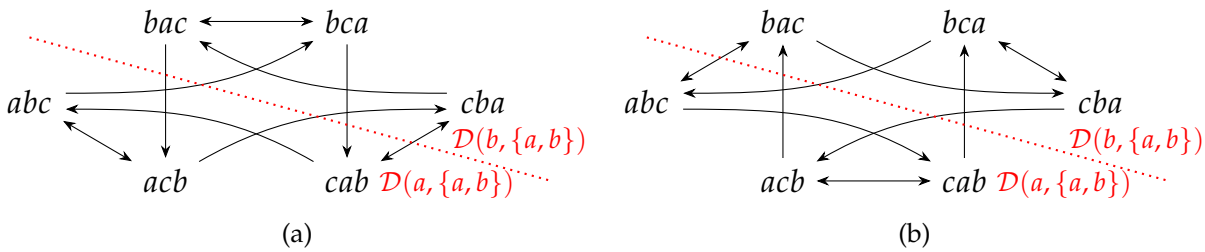


Figure 2: The environments $\langle \mathcal{D}, \mathcal{E}^{\text{drop}} \rangle$ and $\langle \mathcal{D}, \mathcal{E}^{\text{push}} \rangle$ representing drop and push-up strategy-proofness

$\varphi(P')$".

Then how close are k-drop and k-push-up strategy-proofness to strategy-proofness and how does the value of k affect the relationship between them? We can answer these questions by drawing on the results in Section 2.1. Let us begin by making some preliminary observations on drop strategy-proofness. Define an environment $\langle \mathcal{D}, \mathcal{E}^{\text{drop}} \rangle$ as for all $P, P' \in \mathcal{D}$, $(P, P') \in \mathcal{E}^{\text{drop}}$ if and only if P' is a drop strategy for P. Then drop strategy-proofness is just local strategy-proofness associated with $\langle \mathcal{D}, \mathcal{E}^{\text{drop}} \rangle$. Clearly, $\langle \mathcal{D}, \mathcal{E}^{\text{drop}} \rangle$ is not a symmetry environment (e.g., if $A = \{a, b, c\}$, then $(abc, bca) \in \mathcal{E}^{\text{drop}}$ but $(bca, abc) \notin \mathcal{E}^{\text{drop}}$). Similarly, push-up strategy-proofness yields an environment $\langle \mathcal{D}, \mathcal{E}^{\text{push}} \rangle$.

Figure 2 shows $\langle \mathcal{D}, \mathcal{E}^{\text{drop}} \rangle$ and $\langle \mathcal{D}, \mathcal{E}^{\text{push}} \rangle$ in panels (a) and (b), respectively, for the simple case $A = \{a, b, c\}$. One can see that neither is strongly connected because, for instance, $\langle \mathcal{D}(a, \{a, b\}), \mathcal{E}^{\text{drop}}|_{\mathcal{D}(a,\{a,b\})} \rangle$ and $\langle \mathcal{D}(a, \{a, b\}), \mathcal{E}^{\text{push}}|_{\mathcal{D}(a,\{a,b\})} \rangle$ are not connected. Now it follows from Theorem 1 that neither drop nor push-up strategy-proofness implies strategy-proofness (at least for $A = \{a, b, c\}$).

We may subject k-drop and k-push-up strategy-proofness to the same test, for both deterministic and random rules. The environment representing k-drop strategy-proofness is given by $\langle \mathcal{D}, \mathcal{E}^{k\text{-drop}} \rangle$, where for all $P, P' \in \mathcal{D}$, $(P, P') \in \mathcal{E}^{k\text{-drop}}$ if and only if for some $\ell \in \{1, \cdots, k\}$, P' is an ℓ-drop strategy for P. The environment $\langle \mathcal{D}, \mathcal{E}^{k\text{-push}} \rangle$ representing k-push-up strategy-proofness is defined similarly.

It is simple to see that with at most $|A| - 1$ drops (or at most $|A| - 1$ push-ups), we can change an arbitrary preference relation to another. Thus, for $k \geq$

$|A|-1$, k-drop strategy-proofness trivially implies strategy-proofness. However, the implication fails for any $k < |A|-1$, as our next result shows.

Proposition 3. (1) *For deterministic rules, k-drop strategy-proofness implies strategy-proofness if and only if* $k \geq |A|-1$.

(2) *For random rules, k-drop strategy-proofness implies strategy-proofness if and only if* $k \geq |A|-1$.[9]

For this proposition, it is enough to prove the "only if" part of (1) (the "if" part of (1) is trivial and (2) follows from (1)). To show this, we take $P, P' \in \mathcal{D}$ with $a_1\, P\, a_2\, P \cdots P\, a_n$ and $a_n\, P'\, a_1\, P'\, a_2\, P' \cdots P'\, a_{n-1}$ and argue that letting $B \equiv \{a_1, \cdots, a_{n-1}\}$, $P, P' \in \mathcal{D}(a_1, B)$ but for any $k < |A|-1$, there is no path from P to P' in $\langle \mathcal{D}(a_1, B), \mathcal{E}^{k\text{-drop}}|_{\mathcal{D}(a_1,B)} \rangle$, so that $\langle \mathcal{D}, \mathcal{E}^{k\text{-drop}} \rangle$ is not strongly connected.

On the other hand, a counterpart result for k-push-up strategy-proofness turns out to be quite different. As long as $k \geq 2$, k-push-up strategy-proofness implies strategy-proofness for deterministic rules. For random rules, the threshold for k is again $|A|-1$.

Proposition 4. (1) *For deterministic rules, k-push-up strategy-proofness implies strategy-proofness if and only if* $k \geq 2$.[10]

(2) *For random rules, k-push-up strategy-proofness implies strategy-proofness if and only if* $k \geq |A|-1$.

In part (1) of the above proposition, the non-trivial implication is the "if" part. To prove the latter, for each subenvironment $\langle \mathcal{D}(a, B), \mathcal{E}^{k\text{-push}}|_{\mathcal{D}(a,B)} \rangle$ and all $P, P' \in \mathcal{D}(a, B)$, we construct a path from P to P' in $\langle \mathcal{D}(a, B), \mathcal{E}^{k\text{-push}}|_{\mathcal{D}(a,B)} \rangle$ by pushing up at most two alternatives at a time (which is allowed by $k \geq$

[9] Altuntaş et al. (2023) also establish this result, by a different argument.

[10] In a model where agents receive subsets of objects (alternatives), Altuntaş et al. (2023, Proposition 1) show that k-push-up strategy-proofness implies strategy-proofness if and only if $k \geq |A|-1$. At first glance, our part (1) appears to contradict this result but it does not. The two results differ in nature because in our model, the agent receives a single object. When an agent can be assigned multiple objects as in Altuntaş et al. (2023), there is more freedom in constructing rules satisfying k-push-up strategy-proofness but not strategy-proofness than when the agent should be assigned a single object. Consequently, in the setup of Altuntaş et al. (2023), it is more difficult for k-push-up strategy-proofness to imply strategy-proofness. In effect, this is akin to allowing rules to be random and select lotteries. Therefore, the correct counterpart of the above result by Altuntaş et al. (2023) is our part (2).

2). Concerning part (2), the non-trivial implication is the "only if" part. For this, we assume $k < |A| - 1$ and pick $P, P' \in \mathcal{D}$ with $a_1 \, P \, a_2 \, P \cdots P \, a_n$ and $a_2 \, P' \cdots P' \, a_n \, P' \, a_1$. Letting $B \equiv \{a_2, \cdots, a_n\}$ and $C \equiv \{a_2, \cdots, a_{n-1}\}$, while $P, P' \in \mathcal{D}(C, B)$, there is no path from P to P' in $\langle \mathcal{D}(C, B), \mathcal{E}^{k\text{-push}}|_{\mathcal{D}(C,B)} \rangle$, so that $\langle \mathcal{D}, \mathcal{E}^{k\text{-push}} \rangle$ is not hyper-connected.

5 RLGE and the Decomposition of Strategy-proofness

When randomization is allowed, strategy-proofness becomes a very unwieldy property to verify. For each type, one has to check how probabilities for *all* upper contour sets change as various other types are reported. This complication, as well as behavioral motivations, prompts some authors to explore weakenings (different from local strategy-proofness) of strategy-proofness. Chun and Yun (2020) introduce one such property, called upper-contour strategy-proofness. Roy and Sadhukhan (2022) provide conditions on the domain under which strategy-proofness is equivalent to the combination of upper-contour strategy-proofness and some monotonicity properties. In the present application, we derive this result with the aid of Theorem 3 and further show that local strategy-proofness serves as a bridge connecting those incentive properties.

We are interested in the behavior of random rules $\varphi : \mathcal{D} \to \Delta(A)$. As a weakening of strategy-proofness, Chun and Yun (2020) consider an incentive property according to which the agent restricts attention to types that have the same upper contour set as his true type. Below is a formal definition.

Upper-contour Strategy-proofness: For all $P, P' \in \mathcal{D}$ and each $B \subseteq A$ such that B is an upper contour set of both P and P', $\varphi_B(P) \geq \varphi_B(P')$.[11]

We refer the reader to Chun and Yun (2020, p. 671) to see the extent to which upper-contour strategy-proofness is computationally easier to check than strategy-proofness. Upper-contour strategy-proofness is far from being sufficient for strategy-proofness on many domains of practical importance. Thus, Roy and Sadhukhan

[11]Recall that we defined $\varphi_B(P) \equiv \sum_{b \in B} \varphi_b(P)$.

(2022) combine it with monotonicity properties, which are quite appealing in their own right. The first monotonicity property says that if P ranks alternatives a and b consecutively and as we go from P to P', b overtakes just a, with the upper and lower contour sets of $\{a, b\}$ remaining the same, then the probability for a should not increase.

Elementary Monotonicity: For all $P, P' \in \mathcal{D}$ and each partition $\{U, V, \{a\}, \{b\}\}$ of A, where U and V are permitted to be empty, $P = [U, a, b, V]$ and $P' = [U, b, a, V]$ imply $\varphi_a(P) \geq \varphi_a(P')$.

Roy and Sadhukhan (2022) provide a condition on domains under which upper-contour strategy-proofness and elementary monotonicity are equivalent to strategy-proofness. Before introducing it, let us recall that a path $(P_1, \cdots, P_k)$ was defined relative to a given environment $\langle \mathcal{D}, \mathcal{E} \rangle$ (because for each $\ell \in \{1, \cdots, k-1\}$, $(P_\ell, P_{\ell+1}) \in \mathcal{E}$ is required). In this subsection, we consider a sequence $(P_1, \cdots, P_k)$ in $\mathcal{D}$, for which whether P_ℓ is connected to $P_{\ell+1}$ is irrelevant because $\mathcal{E}$ is not specified.[12] Nonetheless, we can still talk about (no-)restoration and overtaking of alternatives along sequences. For instance, we say that for all $x, y \in A$, a sequence $(P_1, \cdots, P_k)$ in $\mathcal{D}$ is **with $\{x, y\}$-restoration** if there are $h, i, j \in \{1, \cdots, k\}$ with $h < i < j$ such that either (i) $x \, P_h \, y$, $y \, P_i \, x$, and $x \, P_j \, y$, or (ii) $y \, P_h \, x$, $x \, P_i \, y$, and $y \, P_j \, x$; and it is **without $\{x, y\}$-restoration** if there is no such (h, i, j). Now we state a domain property introduced by Roy and Sadhukhan (2022).

Definition 8. A domain $\mathcal{D}$ satisfies the **upper-contour no-restoration (UCNR) property** if for all $P, P' \in \mathcal{D}$ and each upper contour set $B \subseteq A$ of P, there is a sequence $(P_1, \cdots, P_k)$ in $\mathcal{D}$, where $P_1 = P$ and $P_k = P'$, such that (i) for each $b \in B$ and each $c \in A \backslash B$, $(P_1, \cdots, P_k)$ is without $\{b, c\}$-restoration; and (ii) for each $h \in \{1, \cdots, k-1\}$, each $b \in B$, and each $c \in A \backslash B$, $b \, P_h \, c$ and $c \, P_{h+1} \, b$ imply that for some disjoint $U, V \subseteq A \backslash \{b, c\}$ with $U \cup V = A \backslash \{b, c\}$, $P_h = [U, b, c, V]$ and $P_{h+1} = [U, c, b, V]$.[13]

A comparison of the UCNR property and Property ULLO is in order. First,

[12] In Roy and Sadhukhan (2002), $(P_1, \cdots, P_k)$ is called a path. In this paper, we reserve the term "path" for sequences consisting of types connected in $\langle \mathcal{D}, \mathcal{E} \rangle$.

[13] One of U and V may be empty.

the UCNR property requires the existence of a sequence while Property ULLO requires the existence of a path. Also, condition (i) of the UCNR property is the same as condition (i) of Property ULLO. Finally, there is a sense in which condition (ii) of the UCNR property is stronger than condition (ii) of Property ULLO (despite their different appearances). One can see the latter point by appending an appropriate set of edges to $\mathcal{D}$, as follows.

Define a set $\boldsymbol{\mathcal{E}}^{\mathbf{UCNR}}$ of edges as follows: for all $P, P' \in \mathcal{D}$, $(P, P') \in \mathcal{E}^{\text{UCNR}}$ if there is $B \subseteq A$ such that letting $\{C_1, \cdots, C_t\}$ be the associated partition for P and P', for each $s \in \{1, \cdots, t\}$, $C_s \cap B \neq \emptyset$ and $C_s \backslash B \neq \emptyset$ imply $|C_s| = 2$. As shown in the proof of Proposition 5 below, the defining condition of $\mathcal{E}^{\text{UCNR}}$, in fact, follows from the UCNR property. Now with $\mathcal{E}^{\text{UCNR}}$, we have an environment $\langle \mathcal{D}, \mathcal{E}^{\text{UCNR}} \rangle$ induced by the UCNR property, on which the notion of local strategy-proofness is well-defined. The environment $\langle \mathcal{D}, \mathcal{E}^{\text{UCNR}} \rangle$ turns out to satisfy Property ULLO. Thus, local strategy-proofness joins the equivalence of strategy-proofness and the combination of upper-contour strategy-proofness and elementary monotonicity.

Proposition 5. *Let $\mathcal{D}$ be a domain satisfying the UCNR property. Define the set $\mathcal{E}^{UCNR}$ of edges as above. Let φ be a random rule on $\langle \mathcal{D}, \mathcal{E}^{UCNR} \rangle$. Then the following statements are equivalent.*[14]

(1) *φ satisfies strategy-proofness.*

(2) *φ satisfies local strategy-proofness.*

(3) *φ satisfies upper-contour strategy-proofness and elementary monotonicity.*

Remark 1. In relation to Roy and Sadhukhan (2022), the significance of Proposition 5 lies in revealing another reason why (1) and (3) of the proposition are equivalent. The equivalence holds because local strategy-proofness can bridge the gap between strategy-proofness and the combination of upper-contour strategy-proofness and elementary monotonicity. In fact, the proof in Cho and Park (2025) of Theorem 3 helps anticipate the latter role played by local strategy-proofness. The proof derives several properties that describe the behavior of

[14]The equivalence of statements (1) and (3) was already proved by Roy and Sadhukhan (2022). The same is true of the equivalence of statements (1) and (3) in Proposition 6 below.

locally strategy-proof rules. Lemma 2 in the proof corresponds to upper-contour strategy-proofness. A simpler version of Lemma 3 gives elementary monotonicity. Thus, the question of whether upper-contour strategy-proofness and elementary monotonicity are together equivalent to strategy-proofness boils down to that of whether LGE holds. A similar comment applies to Proposition 6 below.

$\triangle$

Let us turn to another domain condition that allows a decomposition of strategy-proofness. First, for each $B \subseteq A$, B is a **block in P** if it consists of alternatives ranked consecutively by P; i.e., for some $i, j \in \{1, \cdots, |A|\}$ with $i \leq j$, $B = \{r_k(P) : i \leq k \leq j\}$. For all disjoint blocks $X, Y \subseteq A$, (X, Y) is a **pair of adjacent blocks in P** if for some partition $\{U, V, X, Y\}$ of A, with U and V permitted to be empty, $P = [U, X, Y, V]$. For all $P, P' \in \mathcal{D}$, P and P' are **block adjacent** if there is a collection $\{(X_1, Y_1), \cdots, (X_t, Y_t)\}$ of pairs of adjacent blocks in P such that (i) $\{(Y_1, X_1), \cdots, (Y_t, X_t)\}$ is a collection of pairs of adjacent blocks in P'; and (ii) for all $a, b \in A$, $a\, P\, b$ and $b\, P'\, a$ if and only if for some $s \in \{1, \cdots, t\}$, $a \in X_s$ and $b \in Y_s$. In this case, say that P and P' are **block adjacent with respect to $\{(X_1, Y_1), \cdots, (X_k, Y_k)\}$**. With these concepts, we can define a block counterpart of elementary monotonicity.

Block Monotonicity: For all $P, P' \in \mathcal{D}$ such that P and P' are block adjacent with respect to some collection $\{(X_1, Y_1), \cdots, (X_t, Y_t)\}$, each $s \in \{1, \cdots, t\}$, and each upper contour set $U \subseteq A$ of $P|_{X_s \cup Y_s}$, $\varphi_U(P) \geq \varphi_U(P')$.

Roy and Sadhukhan (2022) identify a domain condition guaranteeing that strategy-proofness is equivalent to the combination of upper-contour strategy-proofness and block monotonicity. Since adjacent blocks are involved in block monotonicity, the domain condition governs the manner in which blocks overtake each other.

Definition 9. A domain $\mathcal{D}$ satisfies the **block restricted upper contour preservation (BRUCP) property** if for all $P, P' \in \mathcal{D}$ and each upper contour set $B \subseteq A$ of P, there is a sequence $(P_1, \cdots, P_k)$ in $\mathcal{D}$, where $P_1 = P$ and $P_k = P'$, satisfying the following: for each $\ell \in \{1, \cdots, k-1\}$, there is a collection $\{(X_1^\ell, Y_1^\ell), \cdots, (X_{t_\ell}^\ell, Y_{t_\ell}^\ell)\}$

of pairs of adjacent blocks in P_ℓ such that (i) P_ℓ and $P_{\ell+1}$ are block adjacent with respect to $\{(X_1^\ell, Y_1^\ell), \cdots, (X_{t_\ell}^\ell, Y_{t_\ell}^\ell)\}$; and (ii) for each $s \in \{1, \cdots, t_\ell\}$, $(X_s^\ell \cup Y_s^\ell) \cap B$ is an upper contour set of $P_\ell|_{X_s^\ell \cup Y_s^\ell}$.

Compared to the UCNR property or Property ULLO, the BRUCP property appears to have no requirement on no-restoration. However, as we show in the proof of Proposition 6 below, it implies the same no-restoration between upper and lower contour sets as in condition (i) of the UCNR property. Moreover, if an appropriate set $\mathcal{E}$ of edges is added to $\mathcal{D}$, the BRUCP property is stronger than Property ULLO.

We establish a result similar to Proposition 5 but involving the BRUCP property. To that end, define a set $\boldsymbol{\mathcal{E}^{\mathbf{BRUCP}}}$ of edges as follows: for all $P, P' \in \mathcal{D}$, $(P, P') \in \mathcal{E}^{\text{BRUCP}}$ if P and P' are block adjacent. The BRUCP property induces an environment $\langle \mathcal{D}, \mathcal{E}^{\text{BRUCP}} \rangle$, which satisfies Property ULLO. Thus, we can juxtapose local strategy-proofness with strategy-proofness, upper-contour strategy-proofness, and block monotonicity, as follows.

Proposition 6. *Let $\mathcal{D}$ be a domain satisfying the BRUCP property. Define the set $\mathcal{E}^{BRUCP}$ of edges as above. Let φ be a random rule on $\langle \mathcal{D}, \mathcal{E}^{BRUCP} \rangle$. Then the following statements are equivalent.*

(1) *φ satisfies strategy-proofness.*

(2) *φ satisfies local strategy-proofness.*

(3) *φ satisfies upper-contour strategy-proofness and block monotonicity.*

Proposition 6 shows that similarly to Remark 1 above, local strategy-proofness serves as a link between strategy-proofness and the combination of upper-contour strategy-proofness and block monotonicity. To see intuitiveness of the link, one may note that the requirements of upper-contour strategy-proofness and block monotonicity represent a simple case of Lemmas 2 and 3 in Appendix A.5, which are consequences of local strategy-proofness.

A Appendix: Proofs

A.1 Proof of Lemma 1

To prove the "if" part, assume that for each ordered pair $(j,k) \in \{1,2\}^2$, $\langle \mathcal{D}, \mathcal{E} \rangle$ is $(\mathcal{D}_j, \mathcal{D}_k)$-strongly connected. Choose $B \subseteq A$ with $|B| \geq 2$ and $a \in B$. Consider $\langle \mathcal{D}(a,B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle$. Let $P, P' \in \mathcal{D}(a,B)$. There is $(j,k) \in \{1,2\}^2$ with $P \in \mathcal{D}_j$ and $P' \in \mathcal{D}_k$. By $(\mathcal{D}_j, \mathcal{D}_k)$-strong connectedness of $\langle \mathcal{D}, \mathcal{E} \rangle$, $\langle \mathcal{D}(a,B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle$ is $(\mathcal{D}_j, \mathcal{D}_k)$-connected. That is, there is a path from P to P' in $\langle \mathcal{D}(a,B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle$.

To prove the "only if" part, assume that $\langle \mathcal{D}, \mathcal{E} \rangle$ is strongly connected. Suppose, by contradiction, that for some $(j,k) \in \{1,2\}^2$, $\langle \mathcal{D}, \mathcal{E} \rangle$ is not $(\mathcal{D}_j, \mathcal{D}_k)$-strongly connected. There exist $B \subseteq A$ with $|B| \geq 2$ and $a \in B$ such that $\langle \mathcal{D}(a,B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle$ is not $(\mathcal{D}_j, \mathcal{D}_k)$-connected; i.e., for some $P \in \mathcal{D}(a,B) \cap \mathcal{D}_j$ and $P' \in \mathcal{D}(a,B) \cap \mathcal{D}_k$, there is no path from P to P' in $\langle \mathcal{D}(a,B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle$. Thus, $\langle \mathcal{D}(a,B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle$ is not connected, a contradiction.

A.2 Proof of Proposition 2

We proceed in two steps.

Step 1: *Showing that $\langle \mathcal{D}, \mathcal{E}^* \rangle$ is $(\mathcal{D}_1, \mathcal{D}_2)$-strongly connected.*

Choose $B \subseteq A$ with $|B| \geq 2$ and $a \in B$. We need to show that $\langle \mathcal{D}(a,B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle$ is $(\mathcal{D}_1, \mathcal{D}_2)$-connected. If $\mathcal{D}(a,B) \subseteq \mathcal{D}_1$ or $\mathcal{D}(a,B) \subseteq \mathcal{D}_2$, then $(\mathcal{D}_1, \mathcal{D}_2)$-connectedness follows trivially. Thus, assume that $\mathcal{D}(a,B) \cap \mathcal{D}_1 \neq \emptyset$ and $\mathcal{D}(a,B) \cap \mathcal{D}_2 \neq \emptyset$. Let $P \in \mathcal{D}(a,B) \cap \mathcal{D}_1$ and $P' \in \mathcal{D}(a,B) \cap \mathcal{D}_2$.

We first assume that for each $(P_1, P_2) \in \mathcal{A}_0$, $M_{(a,B),(P_1,P_2)} = 0$. In view of the algorithm constructing $\mathcal{E}^*$, suppose that in some Step $k \in \{1, \cdots, T\}$, the edge $(P_{k1}, P_{k2}) \in \mathcal{D}_1 \times \mathcal{D}_2$ with $M^k_{(a,B),(P_{k1},P_{k2})} = 1$ is first added to $\mathcal{A}_{k-1}$. In each Step $j \in \{k+1, k+2, \cdots, T\}$, the edge (P_{k1}, P_{k2}) is removed from $\mathcal{A}_{j-1}$ only if there is $(P_1, P_2) \in \mathcal{A}_{j,k-1} \backslash \{(P_{k1}, P_{k2})\}$ with $M^j_{(a,B),(P_1,P_2)} = 1$. This means $P_1, P_2 \in \mathcal{D}(a,B)$. If the edge (P_1, P_2) is also removed in a later step, then there are $P_1' \in \mathcal{D}(a,B) \cap \mathcal{D}_1$ and $P_2' \in \mathcal{D}(a,B) \cap \mathcal{D}_2$ such that the edge (P_1', P_2') is present at the last step (i.e., $(P_1', P_2') \in \mathcal{E}^*$). Since $\langle \mathcal{D}(a,B), \mathcal{E}^*|_{\mathcal{D}(a,B)} \rangle$ is $(\mathcal{D}_1, \mathcal{D}_1)$-connected,

there is a path π from P to P_1' in $\langle \mathcal{D}(a,B), \mathcal{E}^*|_{\mathcal{D}(a,B)} \rangle$. Since $\langle \mathcal{D}(a,B), \mathcal{E}^*|_{\mathcal{D}(a,B)} \rangle$ is $(\mathcal{D}_2, \mathcal{D}_2)$-connected, there is a path π' from P_2' to P' in $\langle \mathcal{D}(a,B), \mathcal{E}^*|_{\mathcal{D}(a,B)} \rangle$. Then concatenating π, (P_1', P_2'), and π', we obtain a path from P to P'.

Next, if there is $(\hat{P}_1, \hat{P}_2) \in \mathcal{A}_0$ with $M_{(a,B),(\hat{P}_1,\hat{P}_2)} = 1$, then by definition, $(\hat{P}_1, \hat{P}_2) \in \mathcal{E} \subseteq \mathcal{E}^*$ and the rest of the argument is the same as the one in the previous paragraph.

Step 2: *Showing that for each set $\mathcal{E}'$ of edges with $\mathcal{E} \subseteq \mathcal{E}' \subsetneq \mathcal{E}^*$, $\langle \mathcal{D}, \mathcal{E}' \rangle$ is not $(\mathcal{D}_1, \mathcal{D}_2)$-strongly connected.*

Let $\mathcal{E}'$ satisfy $\mathcal{E} \subseteq \mathcal{E}' \subsetneq \mathcal{E}^*$. Let $(P_1, P_2) \in \mathcal{E}^* \backslash \mathcal{E}'$. Then $P_1 \in \mathcal{D}_1$ and $P_2 \in \mathcal{D}_2$. For the claim, we may assume that $\mathcal{E}' = \mathcal{E}^* \backslash \{(P_1, P_2)\}$. Consider the algorithm constructing $\mathcal{E}^*$. Assume that in some Step $j \in \{1, \cdots, T\}$, the edge (P_1, P_2) is first added to $\mathcal{A}_{j-1}$, so that $P_1 = P_{j1}$ and $P_2 = P_{j2}$. Clearly, $T \geq j$. If $T = j$, then for each $(\hat{a}, \hat{B}) \in \Phi_T \backslash \Phi_{T-1}$, $\langle \mathcal{D}(\hat{a}, \hat{B}), \mathcal{E}'|_{\mathcal{D}(\hat{a},\hat{B})} \rangle$ is not $(\mathcal{D}_1, \mathcal{D}_2)$-connected. Assume, henceforth, that $T > j$. Since $(P_{j1}, P_{j2}) \in \mathcal{A}_T$, (P_{j1}, P_{j2}) is not removed from $\mathcal{A}_{T,j-1}$ in Substep $T.j$ (i.e., $(P_{j1}, P_{j2}) \in \mathcal{A}_{Tj}$). Then for the vector v constructed in Substep $T.j$, there is $(a, B) \in \Phi_T \subseteq \Phi$ such that $v_{(a,B)} \equiv \sum_{(P_1', P_2') \in \mathcal{A}_{T,j-1} \backslash \{(P_{j1}, P_{j2})\}} M_{(a,B),(P_1',P_2')} = 0$. Thus, $\left\langle \mathcal{D}(a,B), \left(\mathcal{E} \cup \mathcal{A}_{T,j-1} \backslash \{(P_{j1}, P_{j2})\}\right)|_{\mathcal{D}(a,B)} \right\rangle$ is not $(\mathcal{D}_1, \mathcal{D}_2)$-connected. Since no new edge is added after Substep $T.j$, $\mathcal{E}' \subseteq \left(\mathcal{E} \cup \mathcal{A}_{T,j-1} \backslash \{(P_1, P_2)\}\right)$ and $\langle \mathcal{D}(a,B), \mathcal{E}'|_{\mathcal{D}(a,B)} \rangle$ is not $(\mathcal{D}_1, \mathcal{D}_2)$-connected.

A.3 Proof of Proposition 3

(1) The "if" part is trivial and we only prove the "only if" part. For the latter, we prove its contrapositive: if $k < |A| - 1$, then k-drop strategy-proofness does not imply strategy-proofness. By Theorem 1, it is enough to show that $\langle \mathcal{D}, \mathcal{E}^{k\text{-drop}} \rangle$ is not strongly connected. Let us write $A = \{a_1, \cdots, a_n\}$ and choose $P, P' \in \mathcal{D}$ with $a_1 \, P \, a_2 \, P \, \cdots \, P \, a_n$ and $a_n \, P' \, a_1 \, P' \, a_2 \, P' \, \cdots \, P' \, a_{n-1}$. Let $B \equiv \{a_1, \cdots, a_{n-1}\}$. Observe that $P, P' \in \mathcal{D}(a_1, B)$. We show that $\Pi(P, P', \mathcal{E}^{k\text{-drop}}|_{\mathcal{D}(a_1,B)}) = \emptyset$, so that the subenvironment $\langle \mathcal{D}(a_1, B), \mathcal{E}^{k\text{-drop}}|_{\mathcal{D}(a_1,B)} \rangle$ is not connected. Suppose, by contradiction, that there is a path $\pi \equiv (P_1, \cdots, P_m) \in \Pi(P, P', \mathcal{E}^{k\text{-drop}}|_{\mathcal{D}(a_1,B)})$. Since $a_1 \, P \, a_n$ and $a_n \, P' \, a_1$, there is $h \in \{1, \cdots, m-1\}$ such that $a_1 \, P_h \, a_n$ and $a_n \, P_{h+1} \, a_1$.

Since $(P_h, P_{h+1}) \in \mathcal{E}^{k\text{-drop}}$, for some $\ell \in \{1, \cdots, k\}$, P_{h+1} is an ℓ-drop strategy for P_h. This implies that $L(P_{h+1}, a_1)$ contains at most $k-1$ alternatives in A. Since $k-1 < n-2$, there is $a_i \in B$ with $a_i \, P_{h+1} \, a_1$. Thus, $P_{h+1} \notin \mathcal{D}(a_1, B)$, a contradiction.

(2) The "if" part is trivial. The "only if" part also trivially follows from the "only if" part of (1).

A.4 Proof of Proposition 4

Throughout this proof, we write $A = \{a_1, \cdots, a_n\}$.

(1) We already know that push-up strategy-proofness does not imply strategy-proofness. This proves the "only if" part. To prove the "if" part, assume that $k \geq 2$. By Theorem 1, it is enough to show that $\langle \mathcal{D}, \mathcal{E}^{k\text{-push}} \rangle$ is strongly connected. Choose $B \subseteq A$ with $|B| \geq 2$, and $b \in B$. To show that $\langle \mathcal{D}(b, B), \mathcal{E}^{k\text{-push}}|_{\mathcal{D}(b,B)} \rangle$ is connected, let $P, P' \in \mathcal{D}(b, B)$. Without loss of generality, assume that $a_1 \, P \, a_2 \, P \, \cdots \, P \, a_n$. There is a permutation $\sigma : \{1, \cdots, n\} \to \{1, \cdots, n\}$ such that $a_{\sigma(1)} \, P' \, a_{\sigma(2)} \, P' \, \cdots \, P' \, a_{\sigma(n)}$. We construct a path $\pi \equiv (P_1, \cdots, P_m)$ from P to P' in $\langle \mathcal{D}(b, B), \mathcal{E}^{k\text{-push}}|_{\mathcal{D}(b,B)} \rangle$ as follows.

First, starting from $P_1 = P$, push up $a_{\sigma(n-1)}$. If (and only if) $a_{\sigma(n-1)} \in B \backslash \{b\}$, push up b next. Denote the resulting preference relation by P_2. Clearly, $P_2 \in \mathcal{D}(b, B)$.

Second, starting from P_2, push up $a_{\sigma(n-2)}$. If (and only if) $a_{\sigma(n-2)} \in B \backslash \{b\}$, push up b next. Denote the resulting preference relation by P_3. Clearly, $P_3 \in \mathcal{D}(b, B)$.

We may repeat the above process, to push up $a_{\sigma(n-3)}, \cdots, a_{\sigma(1)}$ sequentially and eventually obtain $P_m = P'$. For each $h \in \{1, \cdots, m-1\}$, $(P_h, P_{h+1}) \in \mathcal{E}^{k\text{-push}}$ because we construct P_{h+1} from P_h by pushing up at most two alternatives. Thus, $\pi = (P_1, \cdots, P_m)$ is a path from P to P' in $\langle \mathcal{D}(b, B), \mathcal{E}^{k\text{-push}}|_{\mathcal{D}(b,B)} \rangle$.

(2) The "if" part is trivial. Concerning the "only if" part, we prove its contrapositive: for random rules, if $k < |A| - 1$, then k-push-up strategy-proofness does not imply strategy-proofness. By Theorem 2, it suffices to show that $\langle \mathcal{D}, \mathcal{E}^{k\text{-push}} \rangle$

is not hyper-connected. Choose $P, P' \in \mathcal{D}$ with $a_1 \, P \, a_2 \, P \, \cdots \, P \, a_n$ and $a_2 \, P' \, a_3 \, P' \, \cdots \, P' \, a_n \, P' \, a_1$. Let $B \equiv \{a_2, \cdots, a_n\}$ and $C \equiv \{a_2, \cdots, a_{n-1}\}$. Observe that $P, P' \in \mathcal{D}(C, B)$. We show that $\Pi(P, P', \mathcal{E}^{k\text{-push}}|_{\mathcal{D}(C,B)}) = \emptyset$, so that the subenvironment $\langle \mathcal{D}(C, B), \mathcal{E}^{k\text{-push}}|_{\mathcal{D}(C,B)} \rangle$ is not connected. Suppose, by contradiction, that there is a path $\pi \equiv (P_1, \cdots, P_m) \in \Pi(P, P', \mathcal{E}^{k\text{-push}}|_{\mathcal{D}(C,B)})$. Since $a_1 \, P \, a_n$ and $a_n \, P' \, a_1$, there is $h \in \{1, \cdots, m-1\}$ such that $a_1 \, P_h \, a_n$ and $a_n \, P_{h+1} \, a_1$. Since $(P_h, P_{h+1}) \in \mathcal{E}^{k\text{-push}}$, for some $\ell \in \{1, \cdots, k\}$, P_{h+1} is an ℓ-push-up strategy for P_h. This implies that the upper contour set $U(P_{h+1}, a_n)$ contains at most $k-1$ alternatives in A. Since $k - 1 < n - 2$, there is $a_i \in B$ with $a_n \, P_{h+1} \, a_i$. Thus, $P_{h+1} \notin \mathcal{D}(C, B)$, a contradiction.

A.5 Proof of Proposition 5

Assume that $\mathcal{D}$ is a domain satisfying the UCNR property. The proposition will be proved if we show that (1) implies (2); (2) implies (3); and (3) implies (1). Of these, we prove the last two non-trivial implications.

Before proving the proposition, we present two lemmas from Cho and Park (2025) that play a key role in establishing Theorem 3.

Lemma 2 ("Upper-contour-set (UCS) Lemma"). *Assume that φ is a locally strategy-proof rule. For all $P, P' \in \mathcal{D}$ and each $B \subseteq A$, if B is an upper contour set of both P and P', then $\varphi_B(P) = \varphi_B(P')$.*

Lemma 3 ("Overtaken Lemma"). *Assume that φ is a locally strategy-proof rule. Let $\{U, V, X, Y\}$ be a partition of A, with U and V permitted to be empty. Let $P, P' \in \mathcal{D}$ satisfy $P = [U, X, Y, V]$, $U \, P' \, (Y \cup V)$, and $(U \cup X) \, P' \, V$. Then $\varphi_{U \cup X}(P) \geq \varphi_{U \cup X}(P')$.*

A.5.1 Showing that (2) implies (3)

First, by Lemmas 2 (UCS Lemma) and 3 (Overtaken Lemma), for environments satisfying Property ULLO, each locally strategy-proof random rule satisfies upper-contour strategy-proofness and elementary monotonicity. Thus, it is enough to prove that the environment $\langle \mathcal{D}, \mathcal{E}^{\text{UCNR}} \rangle$ satisfies Property ULLO.

Let $P, P' \in \mathcal{D}$. Let B be an upper contour set of P. By the UCNR property, there is a sequence $(P_1, \cdots, P_k)$ in $\mathcal{D}$, where $P_1 = P$ and $P_k = P'$, such that (i)

for each $b \in B$ and each $c \in A \backslash B$, π is without $\{b, c\}$-restoration; and (ii) for each $\ell \in \{1, \cdots, k-1\}$, each $b \in B$, and each $c \in A \backslash B$, $b \, P_\ell \, c$ and $c \, P_{\ell+1} \, b$ imply $P_\ell = [U, b, c, V]$ and $P_{\ell+1} = [U, c, b, V]$, where $\{U, \{b\}, \{c\}, V\}$ is a partition of A, with U and V permitted to be empty. For each $\ell \in \{1, \cdots, k-1\}$, denote by $\{C_1^\ell, \cdots, C_{t_\ell}^\ell\}$ the associated partition for P_ℓ and $P_{\ell+1}$. We proceed in two steps.

Step 1: *Showing that for each $\ell \in \{1, \cdots, k-1\}$ and each $s \in \{1, \cdots, t_\ell\}$, $C_s^\ell \cap B \neq \emptyset$ and $C_s^\ell \backslash B \neq \emptyset$ imply that for some $b \in B$ and $c \in A \backslash B$, $C_s^\ell = \{b, c\}$, $P_\ell|_{C_s^\ell} = [b, c]$, and $P_{\ell+1}|_{C_s^\ell} = [c, b]$.*

Let $\ell \in \{1, \cdots, k-1\}$ and $s \in \{1, \cdots, t_\ell\}$ satisfy $C_s^\ell \cap B \neq \emptyset$ and $C_s^\ell \backslash B \neq \emptyset$. Write $C_s^\ell \cap B = \{b_1, b_2, \cdots\}$ and $C_s^\ell \backslash B = \{c_1, c_2, \cdots\}$. We proceed in two steps.

Step 1.1: *Showing that if $r_1(P_\ell|_{C_s^\ell}) \in C_s^\ell \backslash B$, say $r_1(P_\ell|_{C_s^\ell}) = c_1$, then we have a contradiction.*

Relabel the alternatives in C_s^ℓ so that $P_\ell|_{C_s^\ell}$ ranks $c_1, \cdots, c_i, b_1, \cdots, b_j$ consecutively (where $i \geq 1$ and $j \geq 1$). Assume that j is the largest integer satisfying

$$P_\ell|_{C_s^\ell} = [c_1, \cdots, c_i, b_1, \cdots, b_j, C_s^\ell \backslash \{c_1, \cdots, c_i, b_1, \cdots, b_j\}].$$

By (i) of the UCNR property, as we go from P_ℓ to $P_{\ell+1}$, no alternative in $\{b_1, \cdots, b_j\}$ overtakes any alternative in $\{c_1, \cdots, c_i\}$. Thus, there is $c_{i+1} \in C_s^\ell \backslash B$ with $b_j \, P_\ell \, c_{i+1}$ (otherwise, $C_s^\ell \backslash \{c_1, \cdots, c_i, b_1, \cdots, b_j\} = \emptyset$ and $P_{\ell+1}|_{C_s^\ell} = [\{c_1, \cdots, c_i\}, \{b_1, \cdots, b_j\}]$, so that $\{C_1^\ell, \cdots, C_{t_\ell}^\ell\}$ is not the finest associated partition for P_ℓ and $P_{\ell+1}$, a contradiction). Without loss of generality, assume that P_ℓ ranks c_{i+1} just below b_j. By (ii) of the UCNR property, c_{i+1} can overtake only b_j in $\{b_1, \cdots, b_j\}$ as we go from P_ℓ to $P_{\ell+1}$.

First, if c_{i+1} does overtake b_j, then by (ii) of the UCNR property,[15]

$$P_\ell|_{C_s^\ell} = [c_1, \cdots, c_i, b_1, \cdots, b_{j-1}, b_j, c_{i+1}, C_s^\ell \backslash \{c_1, \cdots, c_{i+1}, b_1, \cdots, b_j\}];$$
$$P_{\ell+1}|_{C_s^\ell} = [\{c_1, \cdots, c_i\}, \{b_1, \cdots, b_{j-1}\}, c_{i+1}, b_j, C_s^\ell \backslash \{c_1, \cdots, c_{i+1}, b_1, \cdots, b_j\}].$$

Next, assume that c_{i+1} does not overtake b_j. Then b_j is not overtaken by any

[15] If $j = 1$, then $\{b_1, ..., b_{j-1}\}$ will be empty.

alternative and b_j can overtake only those in $\{b_1, \cdots, b_{j-1}\}$. Thus,

$$P_\ell|_{C_s^\ell} = [c_1, \cdots, c_i, b_1, \cdots, b_j, c_{i+1}, C_s^\ell \backslash \{c_1, \cdots, c_{i+1}, b_1, \cdots, b_j\}];$$
$$P_{\ell+1}|_{C_s^\ell} = [\{c_1, \cdots, c_i\}, \{b_1, \cdots, b_j\}, C_s^\ell \backslash \{c_1, \cdots, c_i, b_1, \cdots, b_j\}].$$

In either case, $\{C_1^\ell, \cdots, C_{t_\ell}^\ell\}$ is not the finest associated partition for P_ℓ and $P_{\ell+1}$, a contradiction.

Step 1.2: *Showing that if $r_1(P_\ell|_{C_s^\ell}) \in C_s^\ell \cap B$, say $r_1(P_\ell|_{C_s^\ell}) = b_1$, then $|C_s^\ell| = 2$, with $C_s^\ell = \{b_1, c_1\}$, $P_\ell|_{C_s^\ell} = [b_1, c_1]$, and $P_{\ell+1}|_{C_s^\ell} = [c_1, b_1]$.*

Relabel the alternatives in C_s^ℓ so that $P_\ell|_{C_s^\ell}$ ranks $b_1, \cdots, b_i, c_1, \cdots, c_j$ consecutively (where $i \geq 1$ and $j \geq 1$). Assume that j is the largest integer satisfying

$$P_\ell|_{C_s^\ell} = [b_1, \cdots, b_i, c_1, \cdots, c_j, C_s^\ell \backslash \{b_1, \cdots, b_i, c_1, \cdots, c_j\}]$$

(where we may have $C_s^\ell \backslash \{b_1, \cdots, b_i, c_1, \cdots, c_j\} = \emptyset$ or $j = 1$).

First, assume that $i \geq 2$. By (ii) of the UCNR property, c_1 can overtake only b_i in $\{b_1, \cdots, b_i\}$ as we go from P_ℓ to $P_{\ell+1}$. If c_1 does overtake b_i, then by (ii) of the UCNR property,

$$P_\ell|_{C_s^\ell} = [b_1, \cdots, b_i, c_1, \cdots, c_j, C_s^\ell \backslash \{b_1, \cdots, b_i, c_1, \cdots, c_j\}]$$
$$P_{\ell+1}|_{C_s^\ell} = [\{b_1, \cdots, b_{i-1}\}, c_1, b_i, C_s^\ell \backslash \{b_1, \cdots, b_i, c_1\}].$$

If c_1 does not overtake b_i, then by (i) of the UCNR property, b_i is not overtaken by any alternative and

$$P_\ell|_{C_s^\ell} = [b_1, \cdots, b_i, c_1, \cdots, c_j, C_s^\ell \backslash \{b_1, \cdots, b_i, c_1, \cdots, c_j\}]$$
$$P_{\ell+1}|_{C_s^\ell} = [\{b_1, \cdots, b_i\}, C_s^\ell \backslash \{b_1, \cdots, b_i\}].$$

In either case, $\{C_1^\ell, \cdots, C_{t_\ell}^\ell\}$ is not the finest associated partition for P_ℓ and $P_{\ell+1}$, a contradiction.

The contradiction in the previous paragraph shows $i = 1$. Again to avoid a contradiction to the fact that $\{C_1^\ell, \cdots, C_{t_\ell}^\ell\}$ is the finest associated partition for

P_ℓ and $P_{\ell+1}$, we must have $j = 1$ and $C_s^\ell \backslash \{b_1, \cdots, b_i, c_1, \cdots, c_j\} = \emptyset$. Thus, $C_s^\ell = \{b_1, c_1\}$, $P_\ell|_{C_s^\ell} = [b_1, c_1]$, and $P_{\ell+1}|_{C_s^\ell} = [c_1, b_1]$.

Step 2: *Showing that $\langle \mathcal{D}, \mathcal{E}^{UCNR} \rangle$ satisfies Property ULLO.*

Consider the sequence $(P_1, \cdots, P_k)$ in $\mathcal{D}$. Let $\ell \in \{1, \cdots, k-1\}$ and consider the associated partition $\{C_1^\ell, \cdots, C_{t_\ell}^\ell\}$ for P_ℓ and $P_{\ell+1}$. Let $s \in \{1, \cdots, t_\ell\}$. First, assume that $C_s^\ell \cap B \neq \emptyset$ and $C_s^\ell \backslash B \neq \emptyset$. By Step 1, for some $b \in B$ and $c \in A \backslash B$, $C_s^\ell = \{b, c\}$, $P_\ell|_{C_s^\ell} = [b, c]$, and $P_{\ell+1}|_{C_s^\ell} = [c, b]$. Thus, $(P_\ell, P_{\ell+1}) \in \mathcal{E}^{\text{UCNR}}$ and $P_\ell|_{C_s^\ell} = [C_s^\ell \cap B, C_s^\ell \backslash B]$ (and $P_{\ell+1}|_{C_s^\ell} = [C_s^\ell \backslash B, C_s^\ell \cap B]$). Next, if $C_s^\ell \cap B = \emptyset$ or $C_s^\ell \backslash B = \emptyset$, then trivially, $P_\ell|_{C_s^\ell} = [C_s^\ell \cap B, C_s^\ell \backslash B]$ (and $P_{\ell+1}|_{C_s^\ell} = [C_s^\ell \backslash B, C_s^\ell \cap B]$).

Thus, together with condition (i) of the UCNR property, the argument in the previous paragraph shows that the sequence $(P_1, \cdots, P_k)$ is indeed a path in $\Pi(P, P')$ satisfying the two conditions required by Property ULLO. Thus, $\langle \mathcal{D}, \mathcal{E}^{\text{UCNR}} \rangle$ satisfies Property ULLO.

A.5.2 Showing that (3) implies (1)

Before showing that (3) implies (1), we first note that the same implication as in Lemma 3 (Overtaken Lemma) follows from upper-contour strategy-proofness and elementary monotonicity.

Lemma 4. *Assume that φ satisfies upper-contour strategy-proofness and elementary monotonicity. Let $\{U, V, X, Y\}$ be a partition of A, with U and V permitted to be empty. Let $P, P' \in \mathcal{D}$ satisfy $P = [U, X, Y, V]$, $U\, P'\, (Y \cup V)$, and $(U \cup X)\, P'\, V$. Then $\varphi_{U \cup X}(P) \geq \varphi_{U \cup X}(P')$.*

This lemma can be proved by an argument similar to that in the proof of the Overtaken Lemma. We omit the details.

Now we show that (3) implies (1). Let φ be a random rule satisfying upper-contour strategy-proofness and elementary monotonicity. To show that φ is strategy-proof, let $P, P' \in \mathcal{D}$. Let B be an upper contour set of P. Denote by $\{C_1, \cdots, C_t\}$ the associated partition for P and P'. Let $s \in \{1, \cdots, t\}$ be the smallest integer with $B \subseteq C_1 \cup \cdots \cup C_s$.

First, assume that $B = C_1 \cup \cdots \cup C_s$. Since B is an upper contour set of P' too, upper-contour strategy-proofness implies $\varphi_B(P) \geq \varphi_B(P')$.

Second, assume that $B \subsetneq C_1 \cup \cdots \cup C_s$. By our choice of s, $C_1 \cup \cdots \cup C_{s-1} \subsetneq B \subsetneq C_1 \cup \cdots \cup C_s$. Define $U \equiv C_1 \cup \cdots \cup C_{s-1}$, $V \equiv C_{s+1} \cup \cdots \cup C_t$, $X \equiv C_s \cap B$, and $Y \equiv C_s \backslash B$. Then $P = [U, X, Y, V]$, $U\, P'\, (Y \cup V)$, and $(U \cup X)\, P'\, V$. By Lemma 4, $\varphi_{U \cup X}(P) \geq \varphi_{U \cup X}(P')$. Since $U \cup X = B$, this completes the proof.

A.6 Proof of Proposition 6

Assume that $\mathcal{D}$ is a domain satisfying the BRUCP property. The proposition will be proved if we show that (3) implies (2); (2) implies (1); and (1) implies (3). Of these, we prove the first two non-trivial implications.

A.6.1 Showing that (3) implies (2)

Let φ be a random rule satisfying upper-contour strategy-proofness and block monotonicity. To show that φ is locally strategy-proof, let $P, P' \in \mathcal{D}$ with $(P, P') \in \mathcal{E}^{\text{BRUCP}}$. Let B be an upper contour set of P. Denote by $\{C_1, \cdots, C_v\}$ the associated partition for P and P'. The definition of $\mathcal{E}^{\text{BRUCP}}$ implies that for some collection $\{(X_1, Y_1), \cdots, (X_t, Y_t)\}$ of pairs of adjacent blocks in P, (i) $\{(Y_1, X_1), \cdots, (Y_t, X_t)\}$ is a collection of pairs of adjacent blocks in P'; and (ii) for all $a, b \in A$, $a\, P\, b$ and $b\, P'\, a$ if and only if for some $s \in \{1, \cdots, t\}$, $a \in X_s$ and $b \in Y_s$.

Let $a \in B$ be the alternative bottom-ranked by $P|_B$. Choose $u \in \{1, \cdots, v\}$ with $a \in C_u$.

First, assume that $|C_u| = 1$. Then $B = C_1 \cup \cdots \cup C_u$ is an upper contour set of both P and P'. Thus, upper-contour strategy-proofness implies $\varphi_B(P) \geq \varphi_B(P')$.

Second, assume that $|C_u| \geq 2$. Then for some $s \in \{1, \cdots, t\}$, $C_u = X_s \cup Y_s$. Since $C_1 \cup \cdots \cup C_{u-1}$ is an upper contour set of both P and P', upper-contour strategy-proofness implies $\varphi_{C_1 \cup \cdots \cup C_{u-1}}(P) \geq \varphi_{C_1 \cup \cdots \cup C_{u-1}}(P')$. Also, since $C_u \cap B$ is an upper contour set of $P|_{C_u}$, block monotonicity implies $\varphi_{C_u \cap B}(P) \geq \varphi_{C_u \cap B}(P')$.

Thus,

$$\begin{aligned}\varphi_B(P) &= \varphi_{C_1\cup\cdots\cup C_{u-1}}(P) + \varphi_{C_u\cap B}(P)\\ &\geq \varphi_{C_1\cup\cdots\cup C_{u-1}}(P') + \varphi_{C_u\cap B}(P')\\ &= \varphi_B(P').\end{aligned}$$

A.6.2 Showing that (2) implies (1)

To show that (2) implies (1), it is enough, thanks to Theorem 3, to prove that the environment $\langle \mathcal{D}, \mathcal{E}^{\text{BRUCP}}\rangle$ satisfies Property ULLO. Let $P, P' \in \mathcal{D}$. Let B be an upper contour set of P. By the BRUCP property, there is a sequence $(P_1, \cdots, P_k)$ in $\mathcal{D}$, where $P_1 = P$ and $P_k = P'$, satisfying the following: for each $\ell \in \{1, \cdots, k-1\}$, there is a collection $\{(X_1^\ell, Y_1^\ell), \cdots, (X_{t_\ell}^\ell, Y_{t_\ell}^\ell)\}$ of pairs of adjacent blocks in P_ℓ such that (i) P_ℓ and $P_{\ell+1}$ are block adjacent with respect to $\{(X_1^\ell, Y_1^\ell), \cdots, (X_{t_\ell}^\ell, Y_{t_\ell}^\ell)\}$; and (ii) for each $s \in \{1, \cdots, t_\ell\}$, $(X_s^\ell \cup Y_s^\ell) \cap B$ is an upper contour set of $P_\ell|_{X_s^\ell \cup Y_s^\ell}$. Now we proceed in two steps.

Step 1: *Showing that $\pi \equiv (P_1, \cdots, P_k)$ is a path in $\Pi(P, P')$ satisfying condition (i) of Property ULLO.*

First, by construction, for each $\ell \in \{1, \cdots, k-1\}$, $(P_\ell, P_{\ell+1}) \in \mathcal{E}^{\text{BRUCP}}$. Thus, $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P')$.

Next, we prove that π satisfies condition (i) of Property ULLO. Suppose, by contradiction, that for some $b \in B$ and $c \in A\backslash B$, π is with $\{b, c\}$-restoration. Recalling that $b\, P\, c$, we distinguish two cases.

Case 1: $b\, P'\, c$.

Since π is with $\{b, c\}$-restoration, there is $h \in \{2, \cdots, k-1\}$ with $c\, P_h\, b$ and $b\, P_{h+1}\, c$. Consider P_h and P_{h+1}. Since $c\, P_h\, b$ and $b\, P_{h+1}\, c$, the definition of block adjacency implies that for some $s \in \{1, \cdots, t_h\}$, $c \in X_s^h$ and $b \in Y_s^h$. By the BRUCP property, $(X_s^h \cup Y_s^h) \cap B$ should be an upper contour set of $P_h|_{X_s^h \cup Y_s^h}$. However, even though $c\, P_h\, b$, $(X_s^h \cup Y_s^h) \cap B$ contains b but not c, a contradiction.

Case 2: $c\, P'\, b$.

Since π is with $\{b, c\}$-restoration, there is $h \in \{2, \cdots, k-2\}$ with $c\, P_h\, b$ and

$b\, P_{h+1}\, c$. Now we can apply the argument in Case 1 to P_h and P_{h+1} and get a similar contradiction.

Step 2: *Showing that π satisfies condition (ii) of Property ULLO.*

For each $\ell \in \{1, \cdots, k-1\}$, denote by $\{C_1^\ell, \cdots, C_{v_\ell}^\ell\}$ the associated partition for $(P_\ell, P_{\ell+1})$. Let $\ell \in \{1, \cdots, k-1\}$. By the definition of block adjacency, for each $u \in \{1, \cdots, v_\ell\}$, $|C_u^\ell| \geq 2$ implies that for some $s \in \{1, \cdots, t_\ell\}$, $C_u^\ell = X_s^\ell \cup Y_s^\ell$ and that (by (ii) of the BRUCP property) $P_\ell|_{C_u^\ell} = [C_u^\ell \cap B, C_u^\ell \backslash B]$.

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